

Config 10 (0.05|0.9|1|1|0.52|0.5| 0.7| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.700$

effect $\pi_t = 0.740$ ($= \pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.pstest)

Power (P reject ≥ 1 | H1) : 0.9012

FWER (P reject ≥ 1 | H0) : 0.0517

Max sample size (planned) : 4314

ESS under H1 : 4314.0

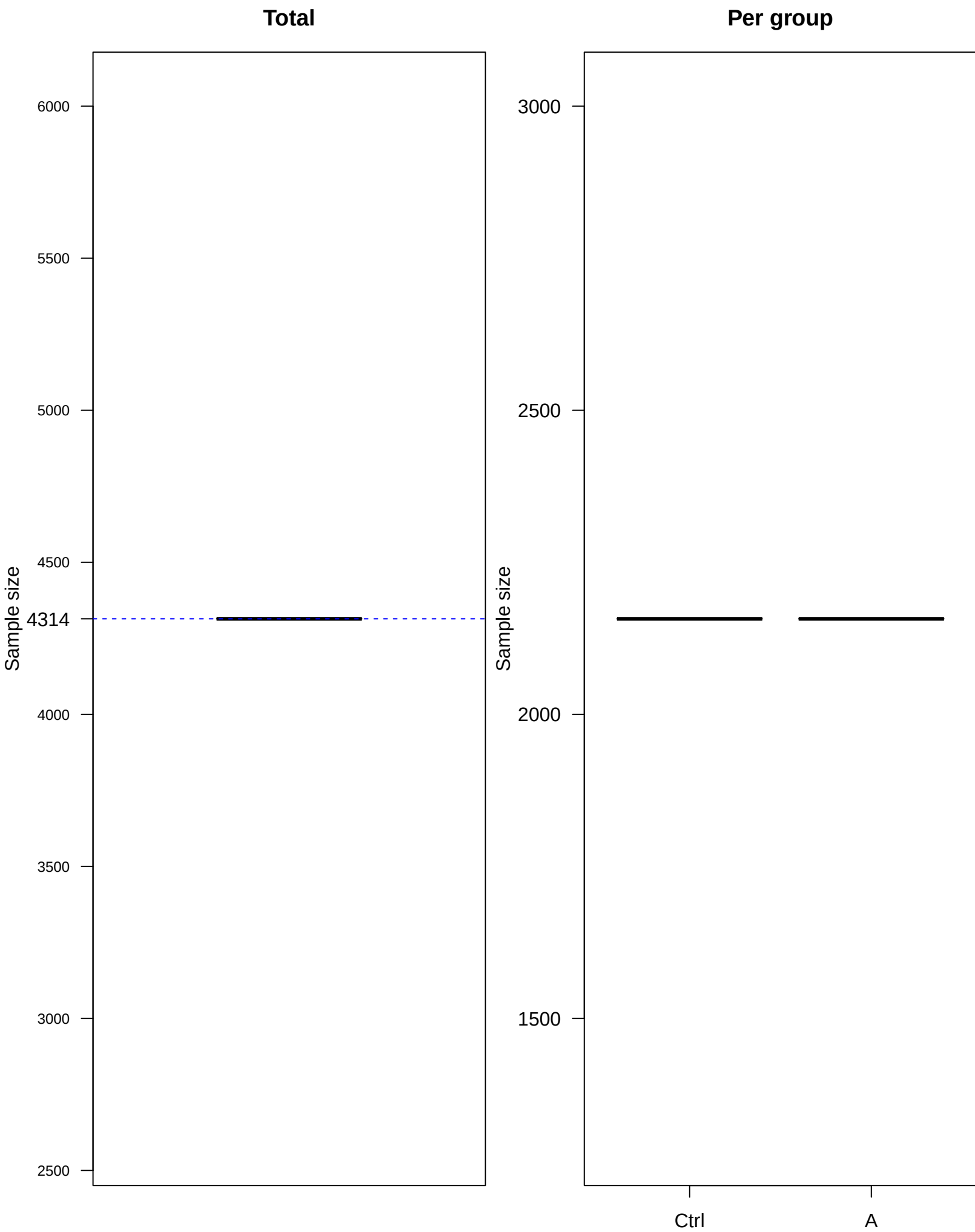
ESS under H0 : 4314.0

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(δ/σ); residual gap from target is small-sample discreteness.

Config 10 – sample size (H1) [binary]



Config 12 (0.05|0.9|1|1|0.52|0.5| 0.9| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.900

effect pi_t = 0.940 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9048

FWER (P reject ≥ 1 | H0) : 0.0489

Max sample size (planned) : 1572

ESS under H1 : 1572.0

ESS under H0 : 1572.0

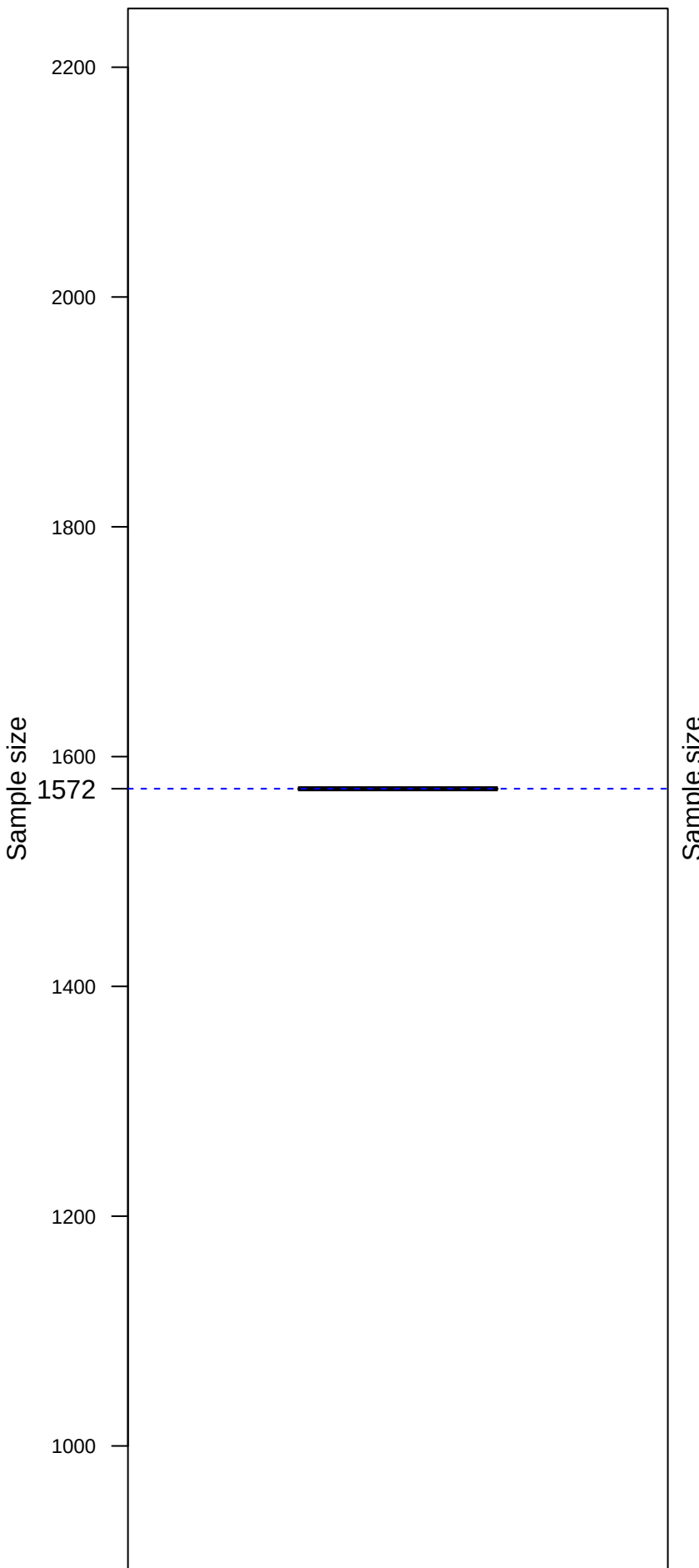
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

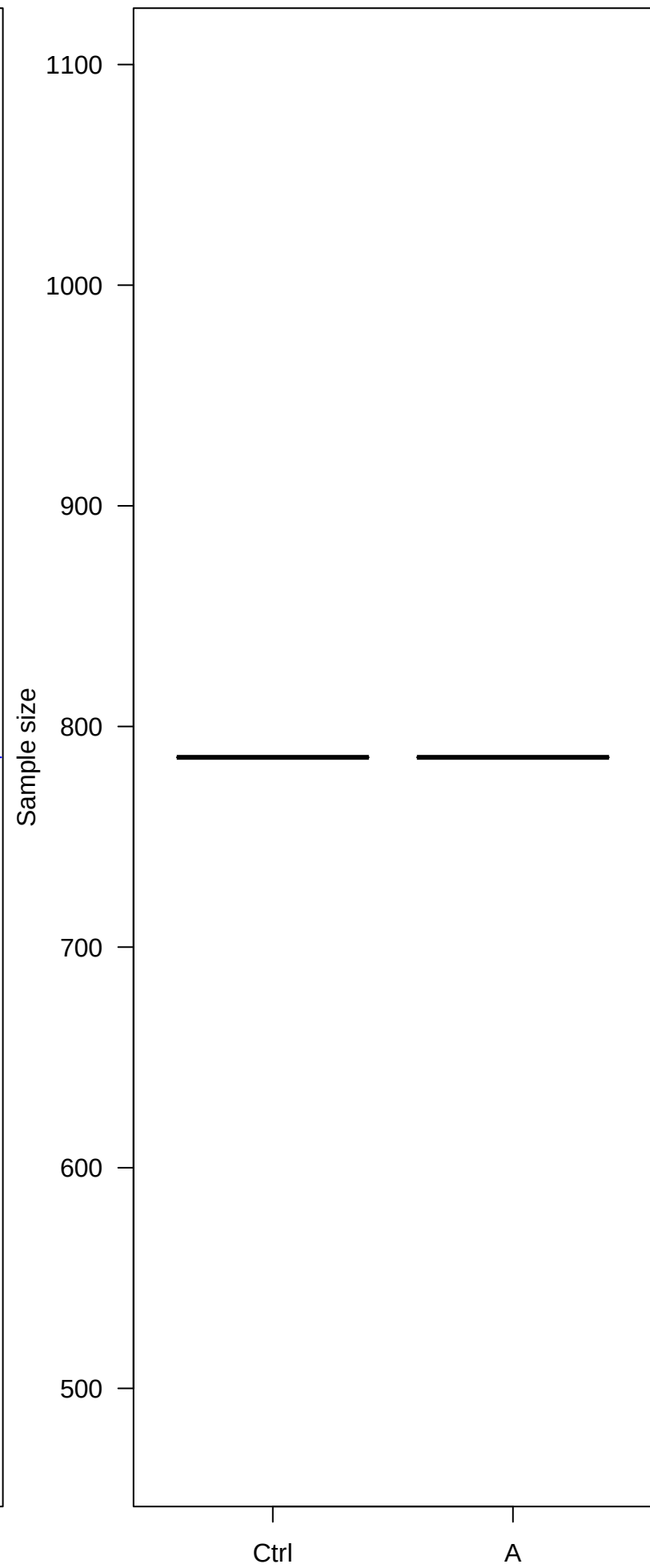
(delta/sd); residual gap from target is small-sample discreteness.

Config 12 – sample size (H1) [binary]

Total



Per group



Config 20 (0.05|0.9|1|1|0.55|0.5| 0.5| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.55

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9023

FWER (P reject ≥ 1 | H0) : 0.0532

Max sample size (planned) : 846

ESS under H1 : 846.0

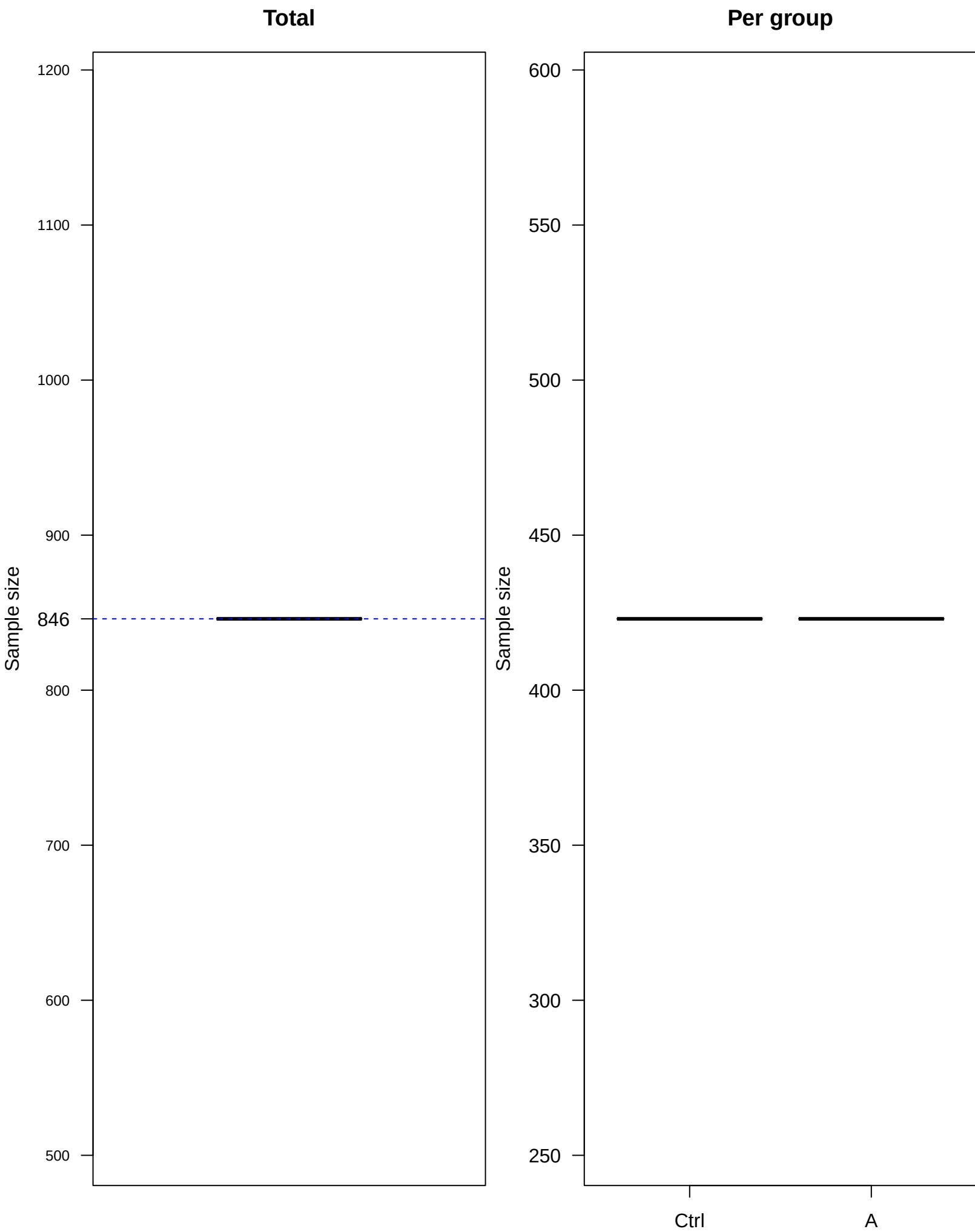
ESS under H0 : 846.0

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 20 – sample size (H1) [binary]



Config 34 (0.05|0.9|1|1| 0.6|0.5| 0.7| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.6

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.700

effect pi_t = 0.900 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9126

FWER (P reject ≥ 1 | H0) : 0.0524

Max sample size (planned) : 134

ESS under H1 : 134.0

ESS under H0 : 134.0

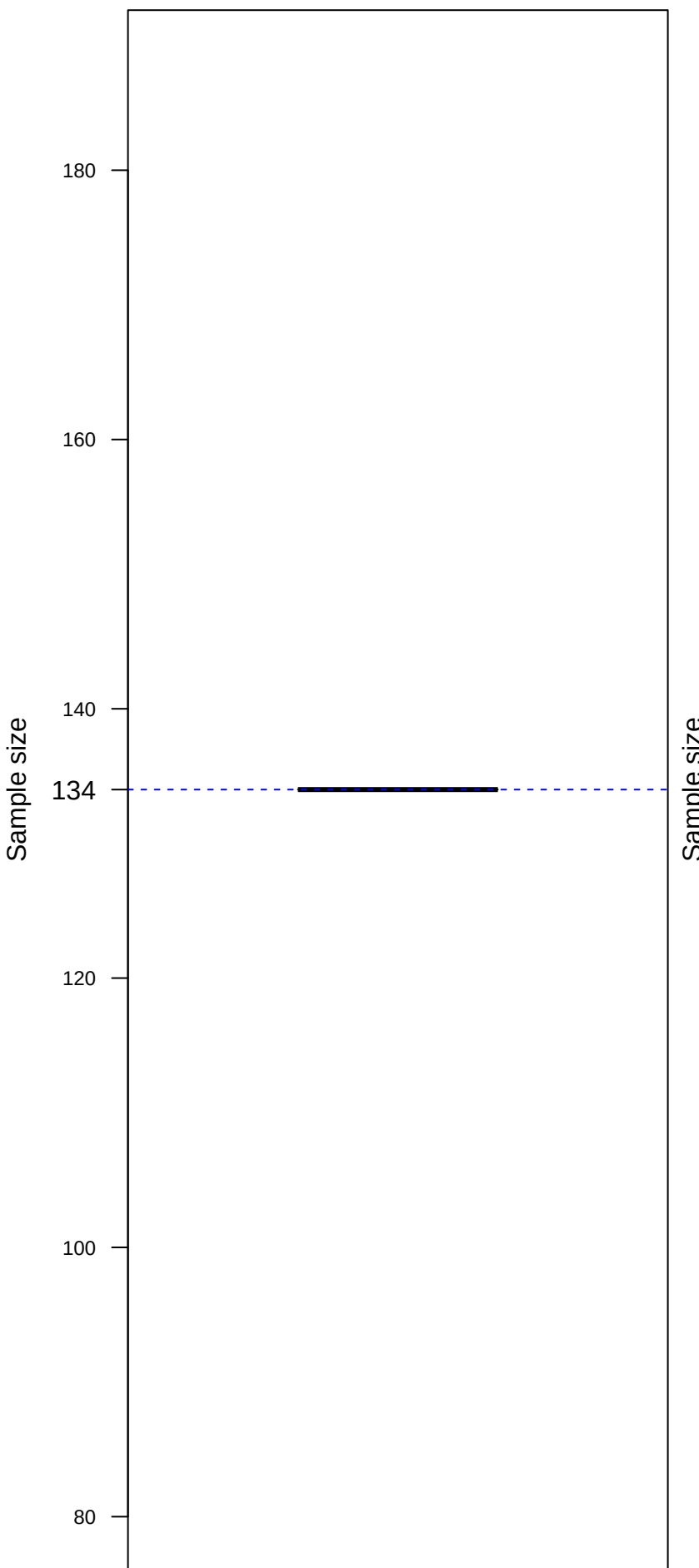
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

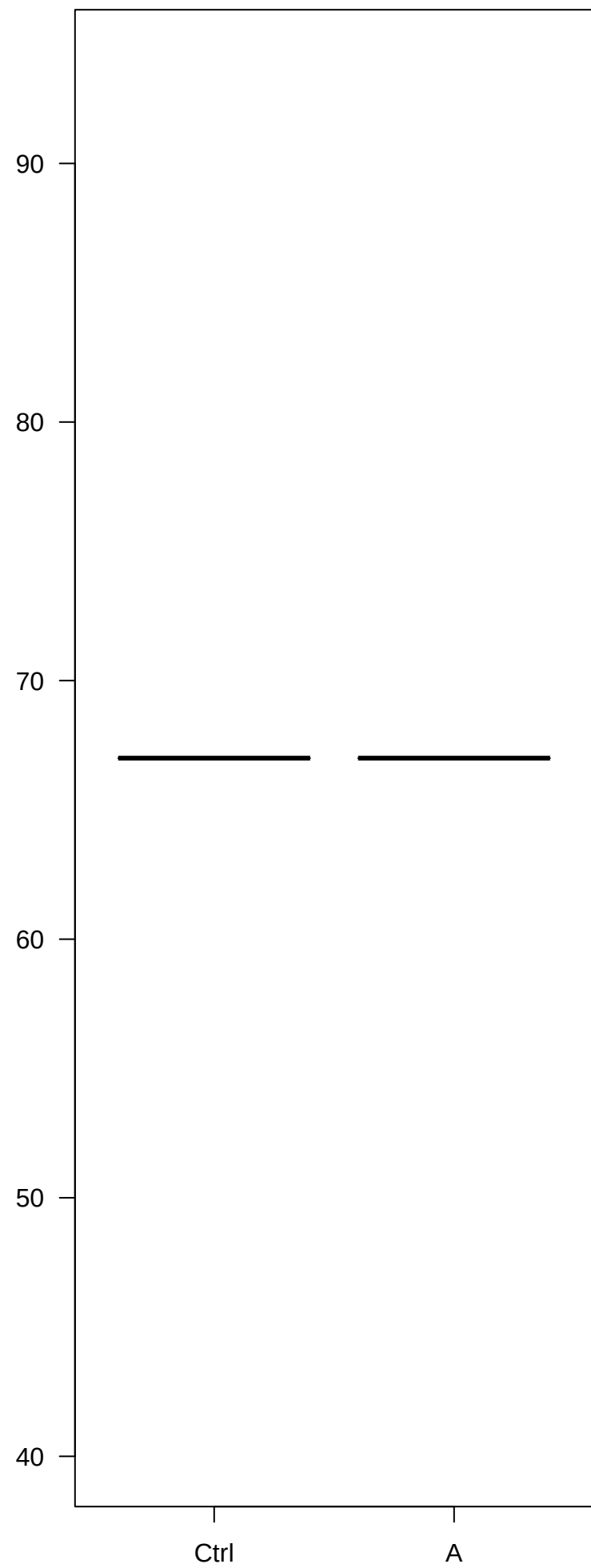
(delta/sd); residual gap from target is small-sample discreteness.

Config 34 – sample size (H1) [binary]

Total



Per group



Config 38 (0.05|0.9|1|1|0.65|0.5|0.05| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.350$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9314

FWER (P reject ≥ 1 | H0) : 0.0450

Max sample size (planned) : 58

ESS under H1 : 58.0

ESS under H0 : 58.0

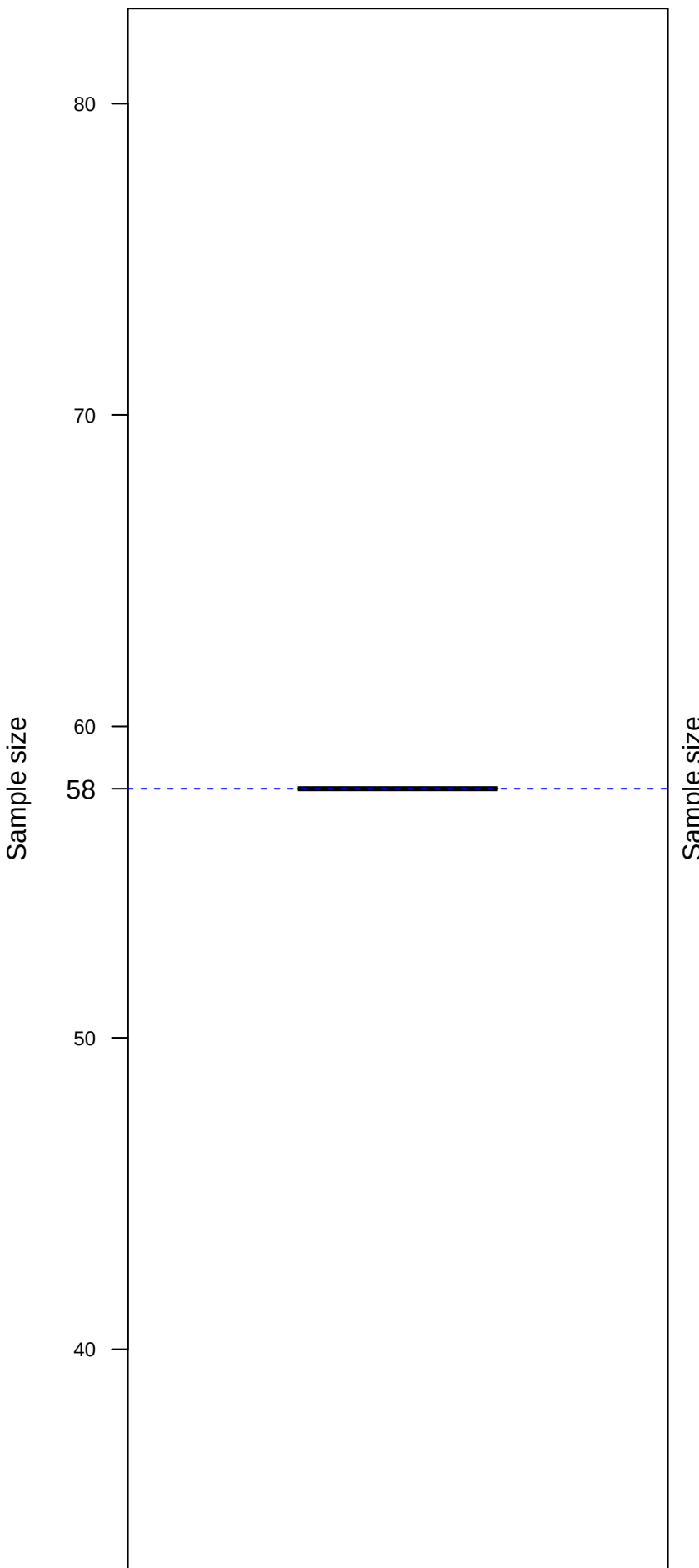
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

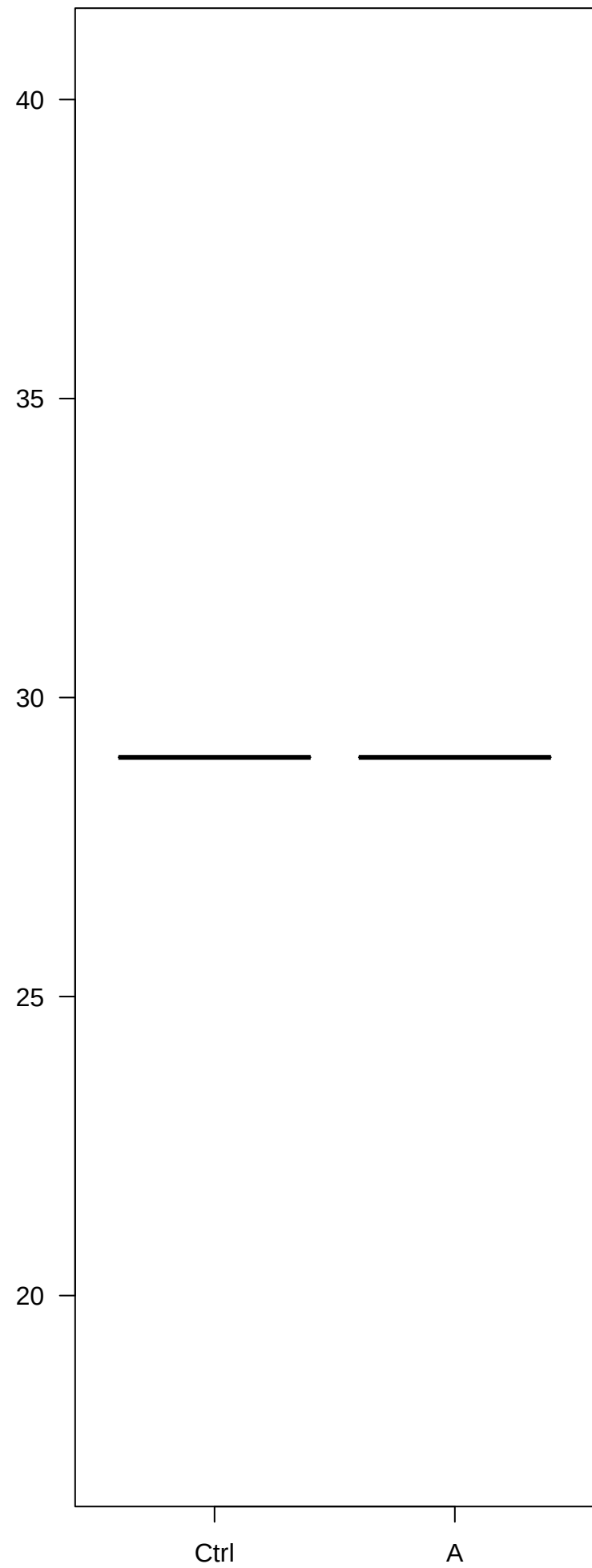
(delta/sd); residual gap from target is small-sample discreteness.

Config 38 – sample size (H1) [binary]

Total



Per group



Config 42 (0.05|0.9|1|1|0.65|0.5| 0.3| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9157

FWER (P reject ≥ 1 | H0) : 0.0508

Max sample size (planned) : 92

ESS under H1 : 92.0

ESS under H0 : 92.0

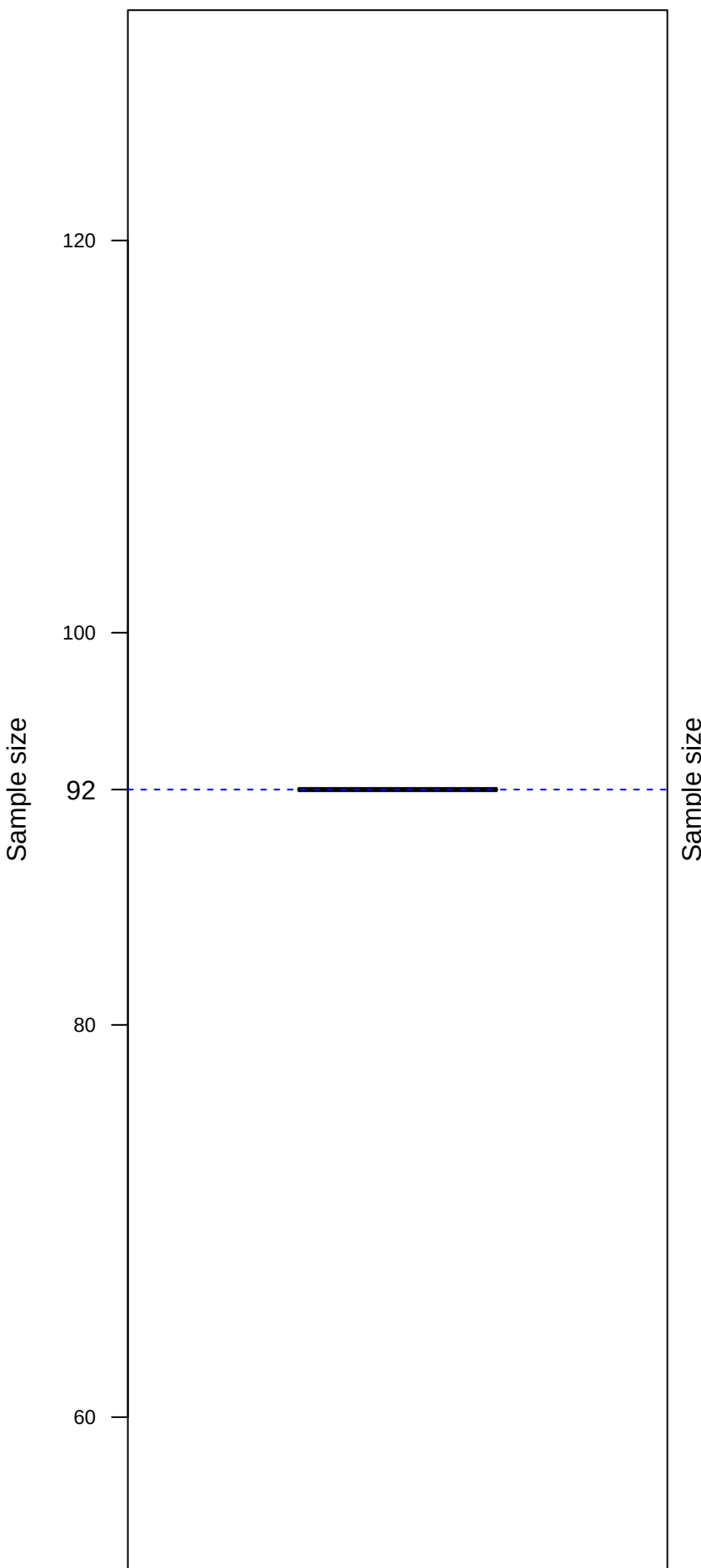
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

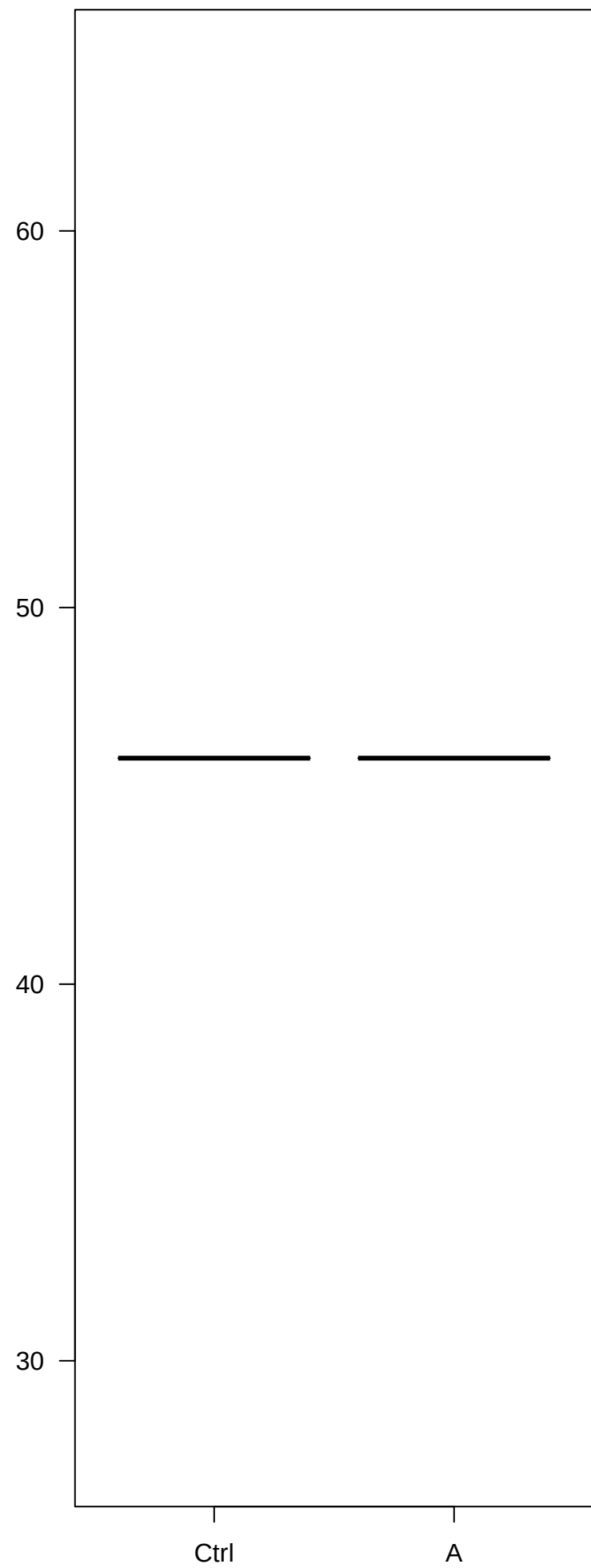
(delta/sd); residual gap from target is small-sample discreteness.

Config 42 – sample size (H1) [binary]

Total



Per group



Config 44 (0.05|0.9|1|1|0.65|0.5| 0.5| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9026

FWER (P reject ≥ 1 | H0) : 0.0544

Max sample size (planned) : 84

ESS under H1 : 84.0

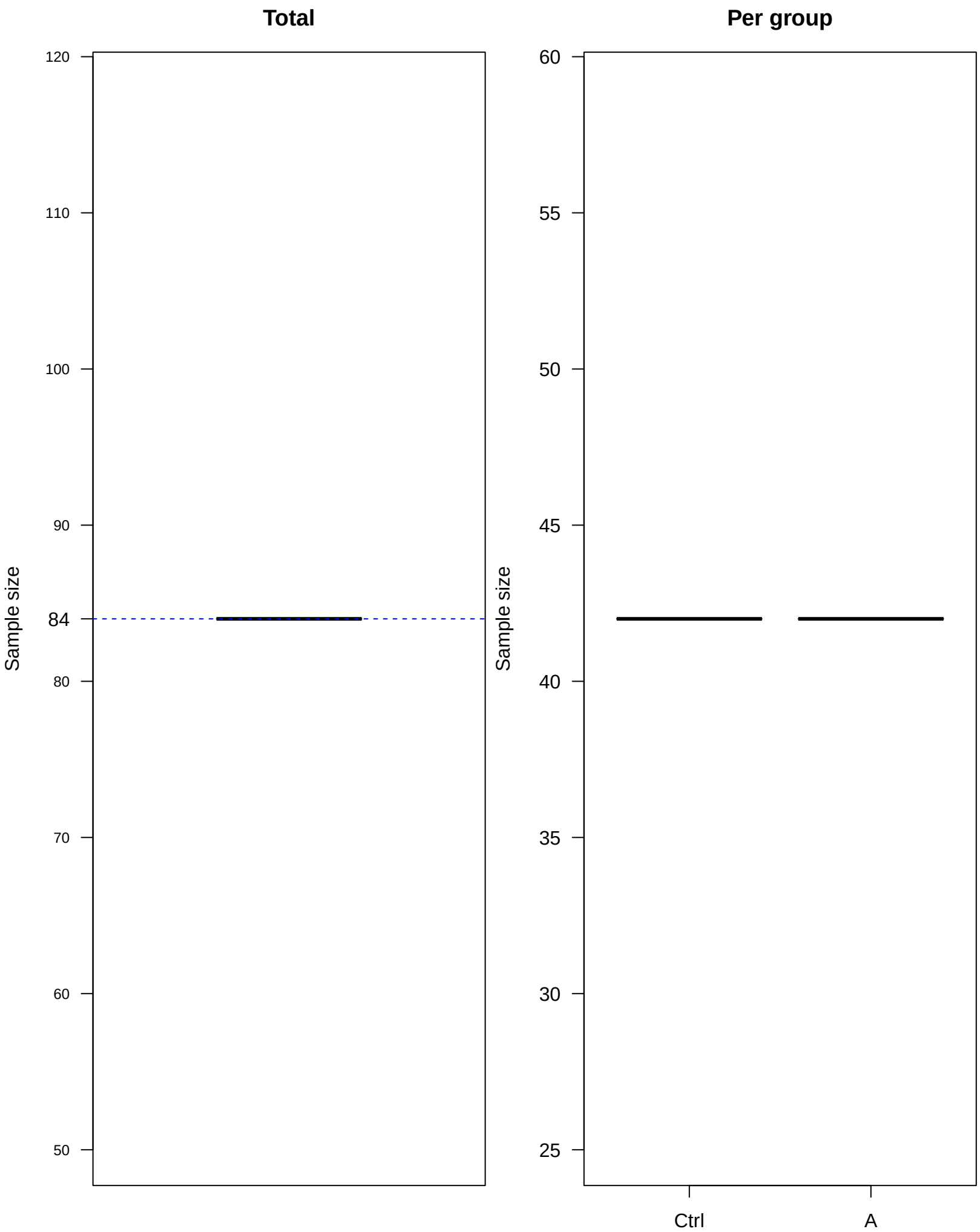
ESS under H0 : 84.0

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 44 – sample size (H1) [binary]



Config 54 (0.05|0.9|1|1|0.75|0.5| 0.3| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9195

FWER (P reject ≥ 1 | H0) : 0.0473

Max sample size (planned) : 32

ESS under H1 : 32.0

ESS under H0 : 32.0

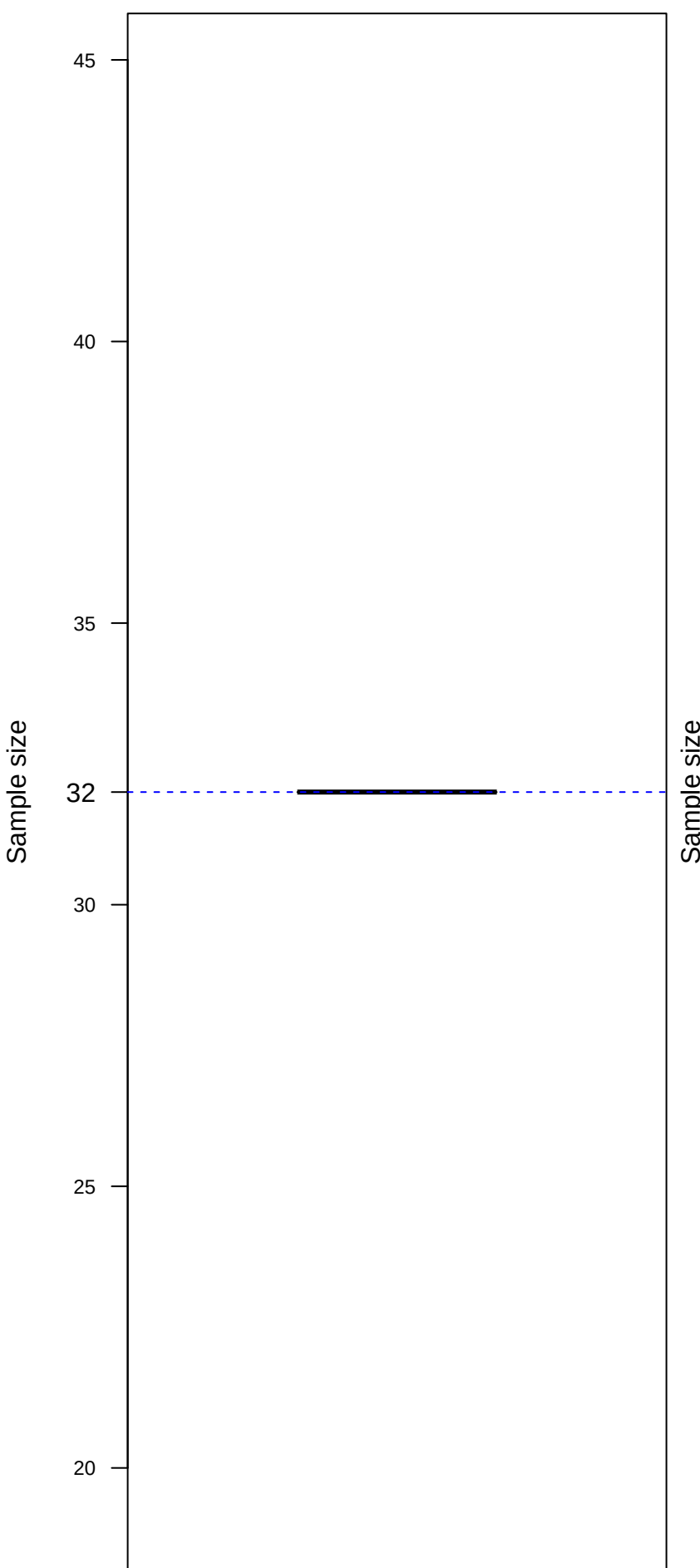
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

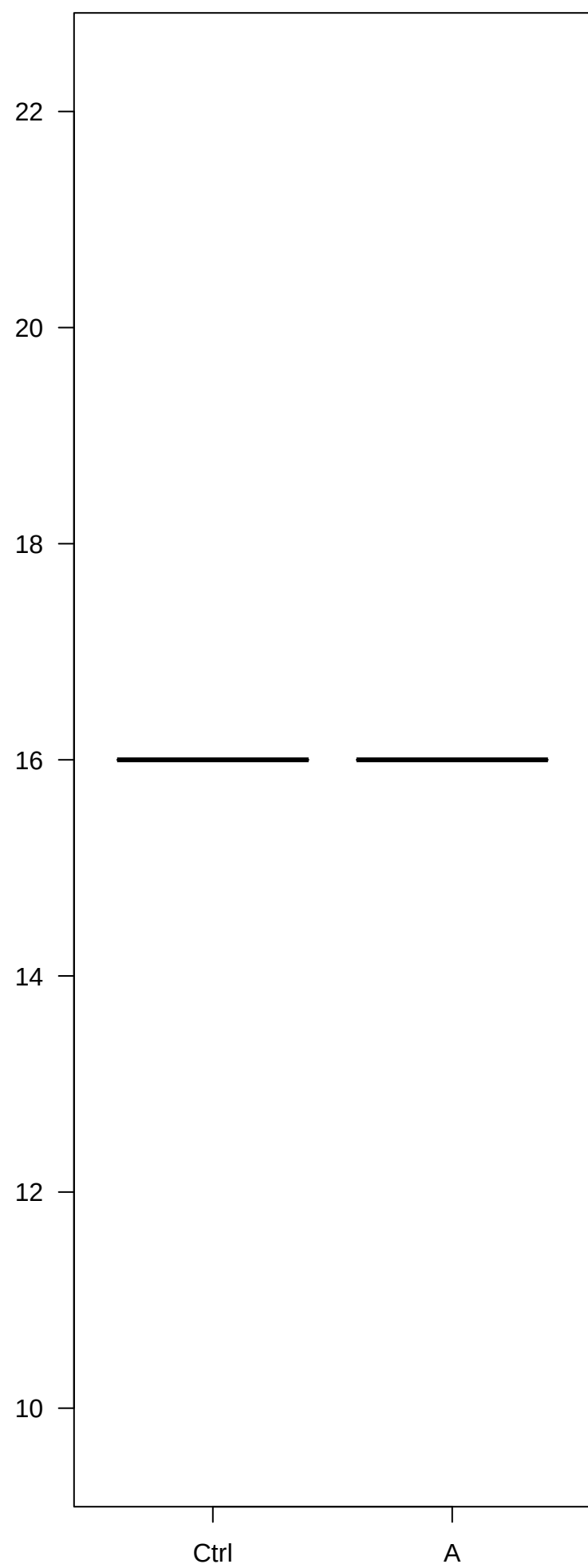
(delta/sd); residual gap from target is small-sample discreteness.

Config 54 – sample size (H1) [binary]

Total



Per group



Config 64 (0.05|0.9|1|1|0.85|0.5| 0.1| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.85

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.100

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9230

FWER (P reject ≥ 1 | H0) : 0.0134

Max sample size (planned) : 14

ESS under H1 : 14.0

ESS under H0 : 14.0

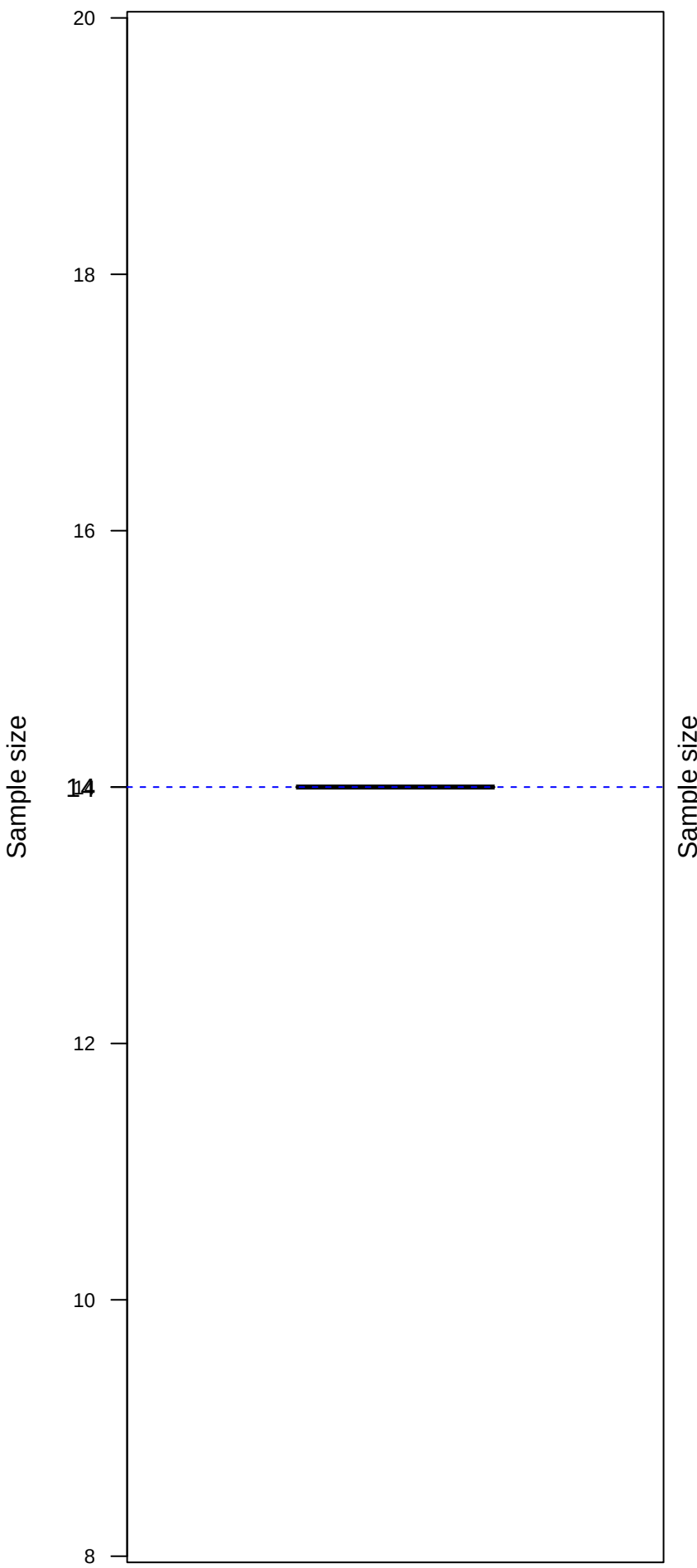
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

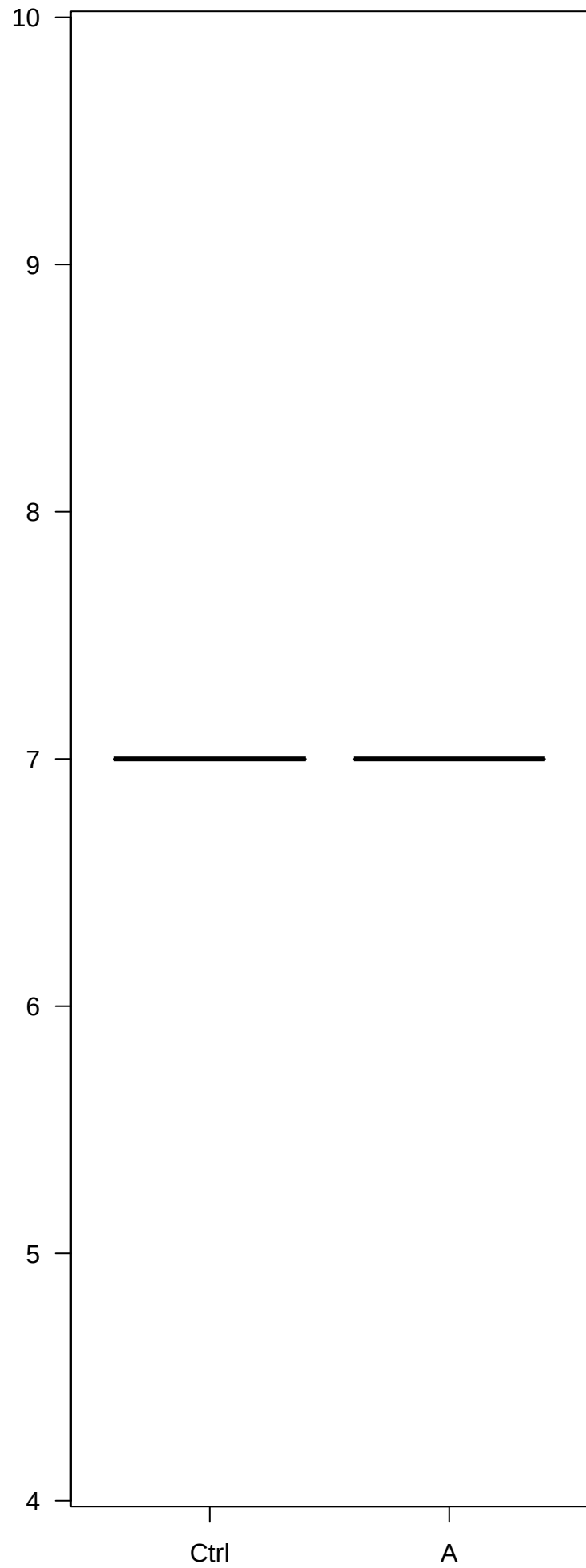
(delta/sd); residual gap from target is small-sample discreteness.

Config 64 – sample size (H1) [binary]

Total



Per group



Config 74 (0.05|0.9|1|1|0.95|0.5|0.05| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 1

p (target win-prob) : 0.95

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.050

effect pi_t = 0.950 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9673

FWER (P reject ≥ 1 | H0) : 0.0064

Max sample size (planned) : 6

ESS under H1 : 6.0

ESS under H0 : 6.0

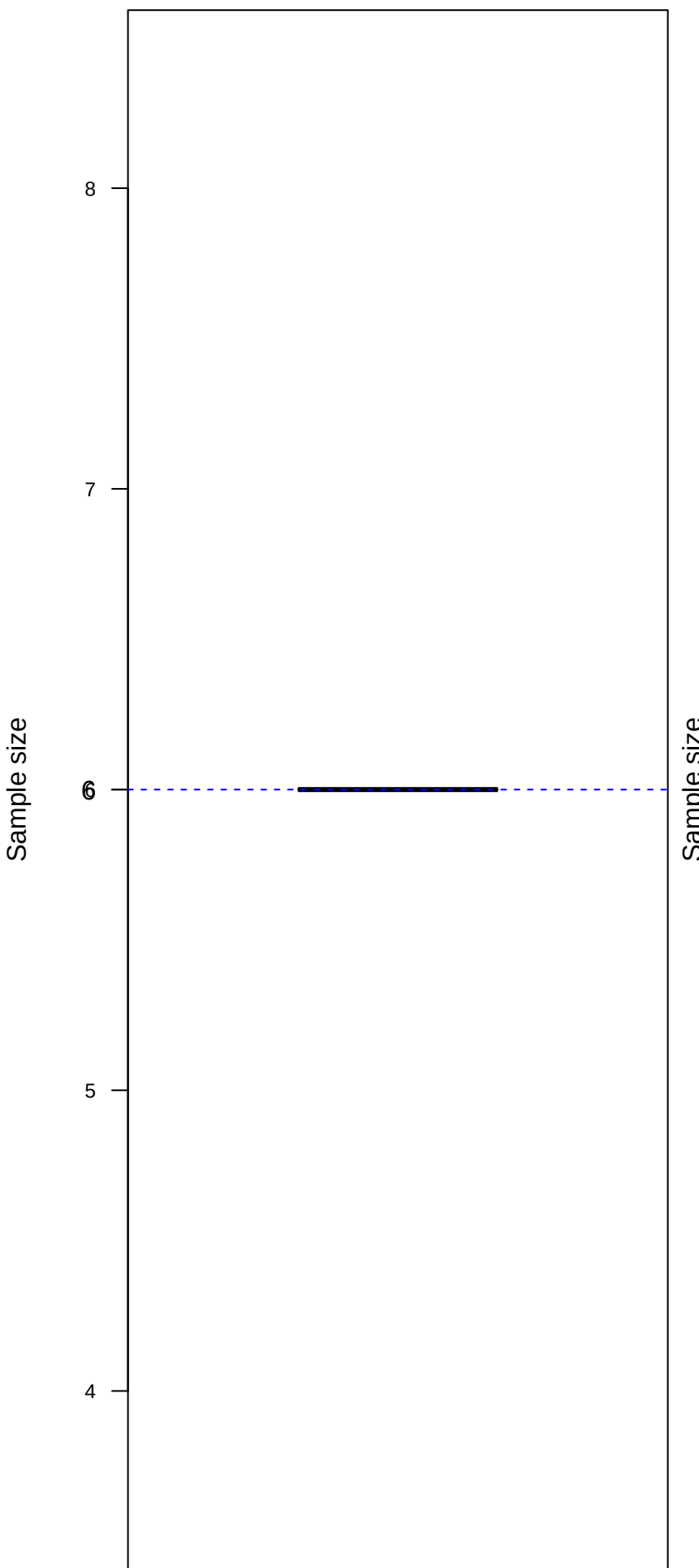
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

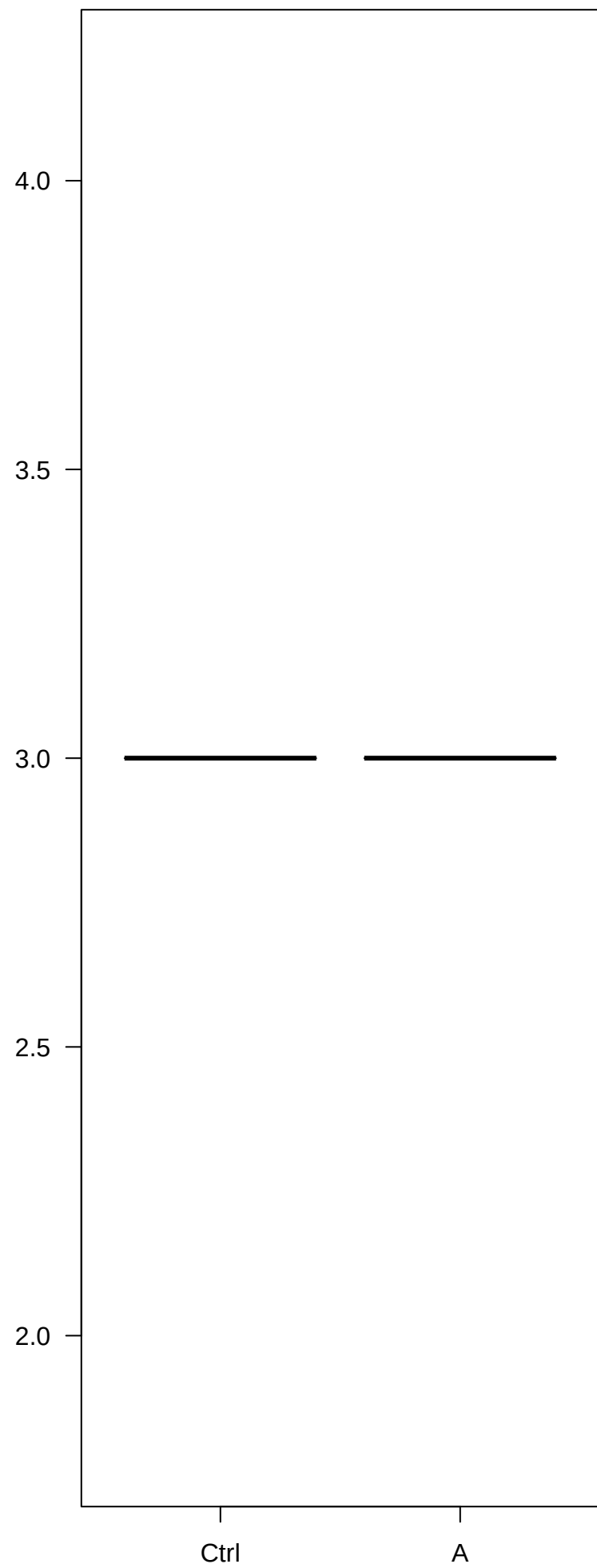
(delta/sd); residual gap from target is small-sample discreteness.

Config 74 – sample size (H1) [binary]

Total



Per group



Config 94 (0.05|0.9|1|2|0.52|0.5| 0.7| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.700

effect pi_t = 0.740 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9000

FWER (P reject ≥ 1 | H0) : 0.0455

Max sample size (planned) : 4904

ESS under H1 : 3708.7

ESS under H0 : 3100.3

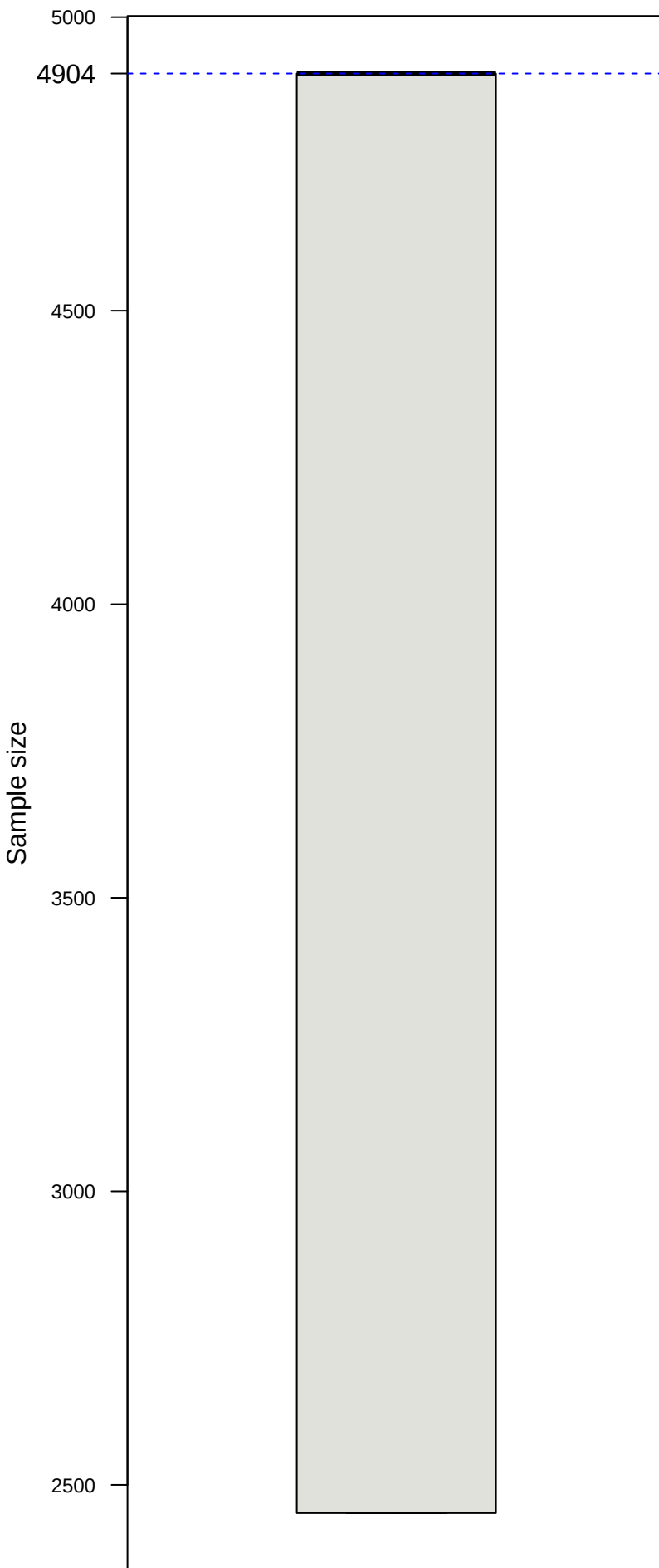
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

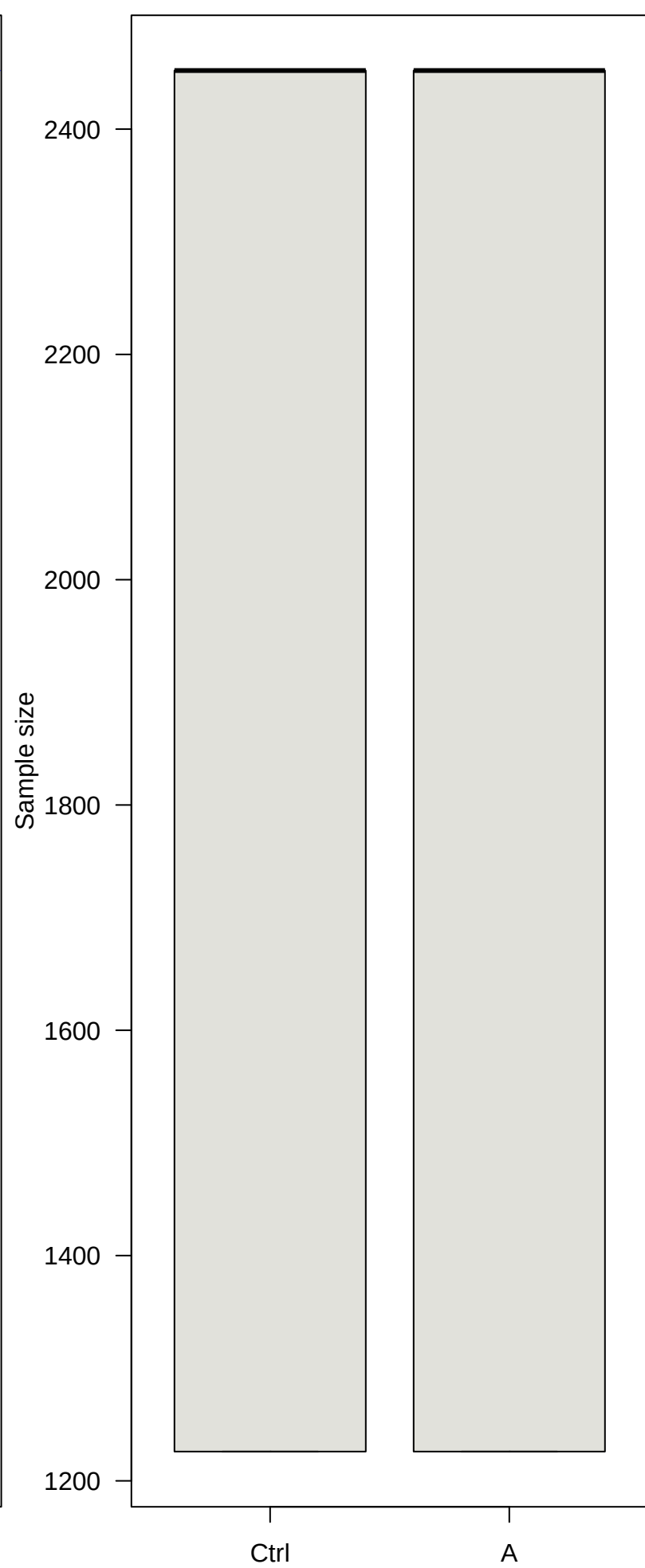
(delta/sd); residual gap from target is small-sample discreteness.

Config 94 – sample size (H1) [binary]

Total



Per group



Config 96 (0.05|0.9|1|2|0.52|0.5| 0.9| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.900

effect pi_t = 0.940 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9030

FWER (P reject ≥ 1 | H0) : 0.0437

Max sample size (planned) : 1788

ESS under H1 : 1360.0

ESS under H0 : 1128.8

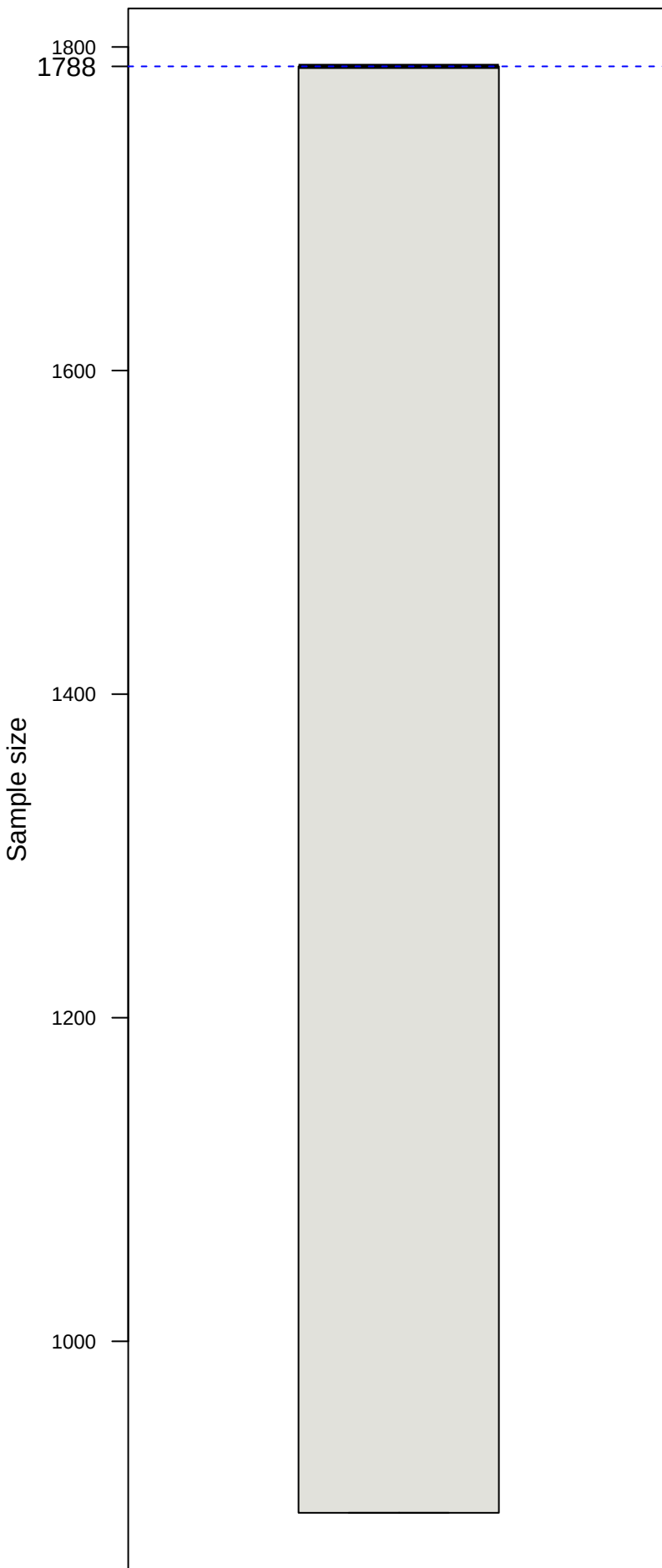
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

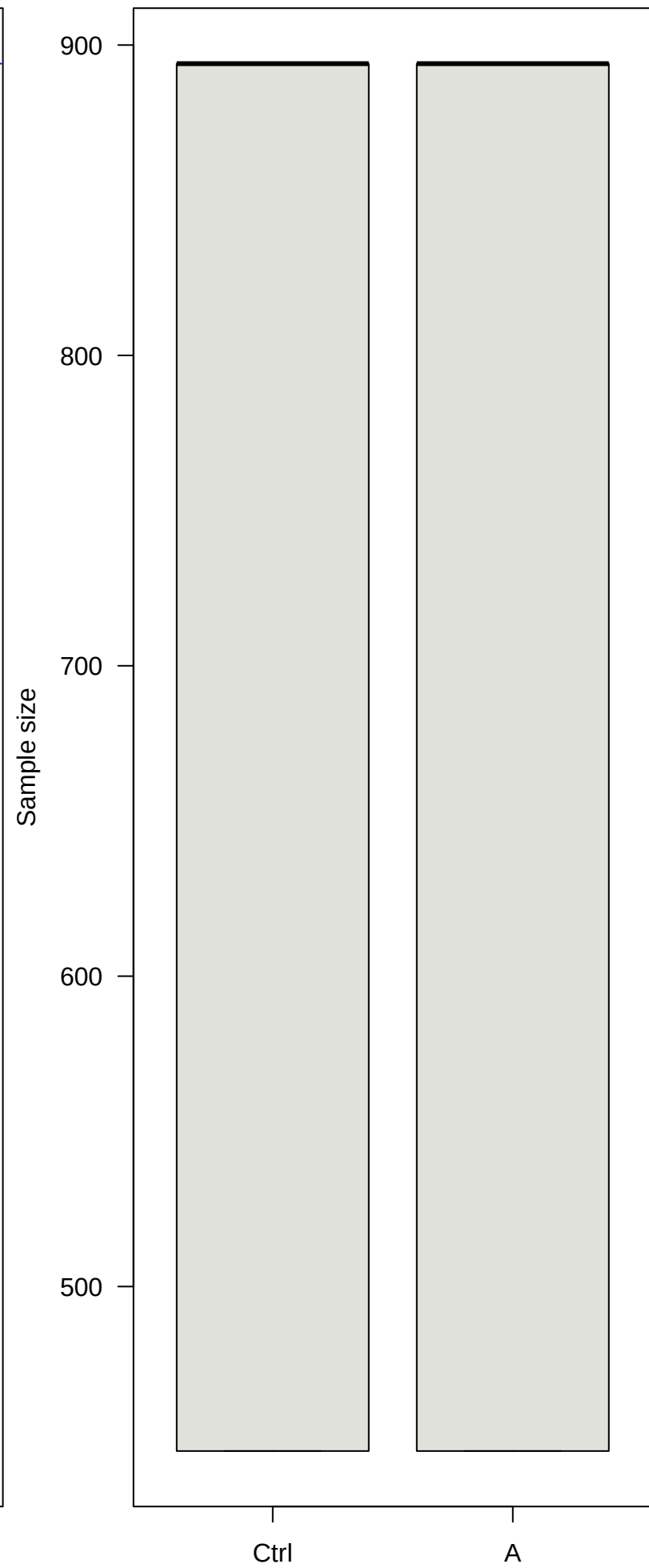
(delta/sd); residual gap from target is small-sample discreteness.

Config 96 – sample size (H1) [binary]

Total



Per group



Config 104 (0.05|0.9|1|2|0.55|0.5| 0.5| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.55

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9025

FWER (P reject ≥ 1 | H0) : 0.0476

Max sample size (planned) : 960

ESS under H1 : 723.0

ESS under H0 : 607.9

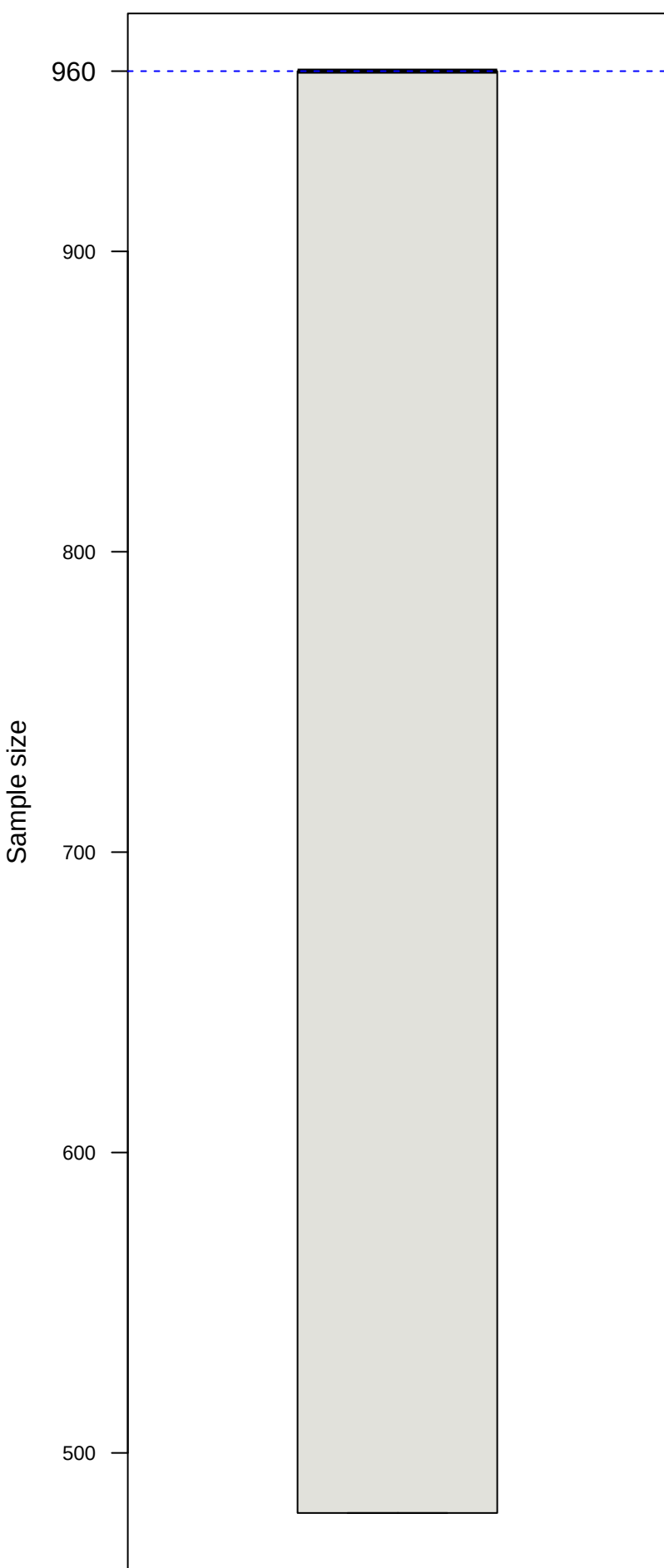
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

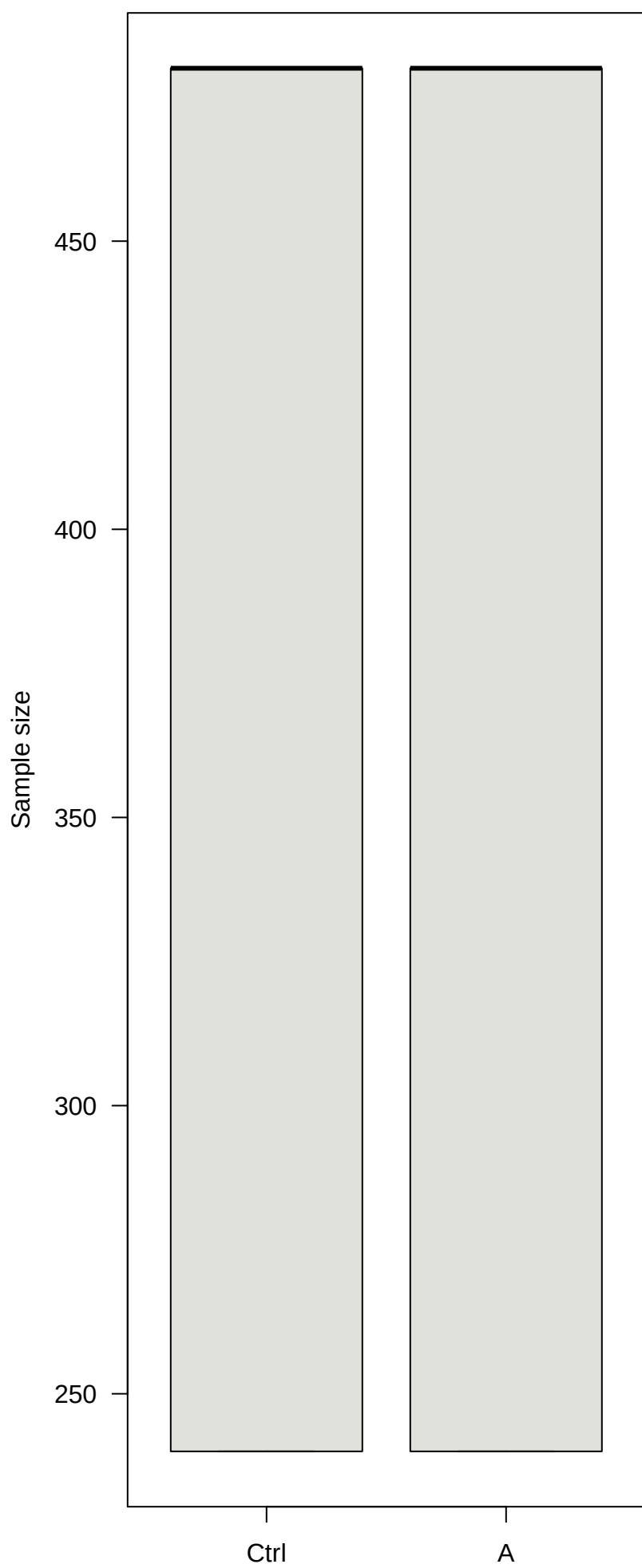
(delta/sd); residual gap from target is small-sample discreteness.

Config 104 – sample size (H1) [binary]

Total



Per group



Config 118 (0.05|0.9|1|2| 0.6|0.5| 0.7| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.6

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.700

effect pi_t = 0.900 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9065

FWER (P reject ≥ 1 | H0) : 0.0452

Max sample size (planned) : 152

ESS under H1 : 116.1

ESS under H0 : 95.8

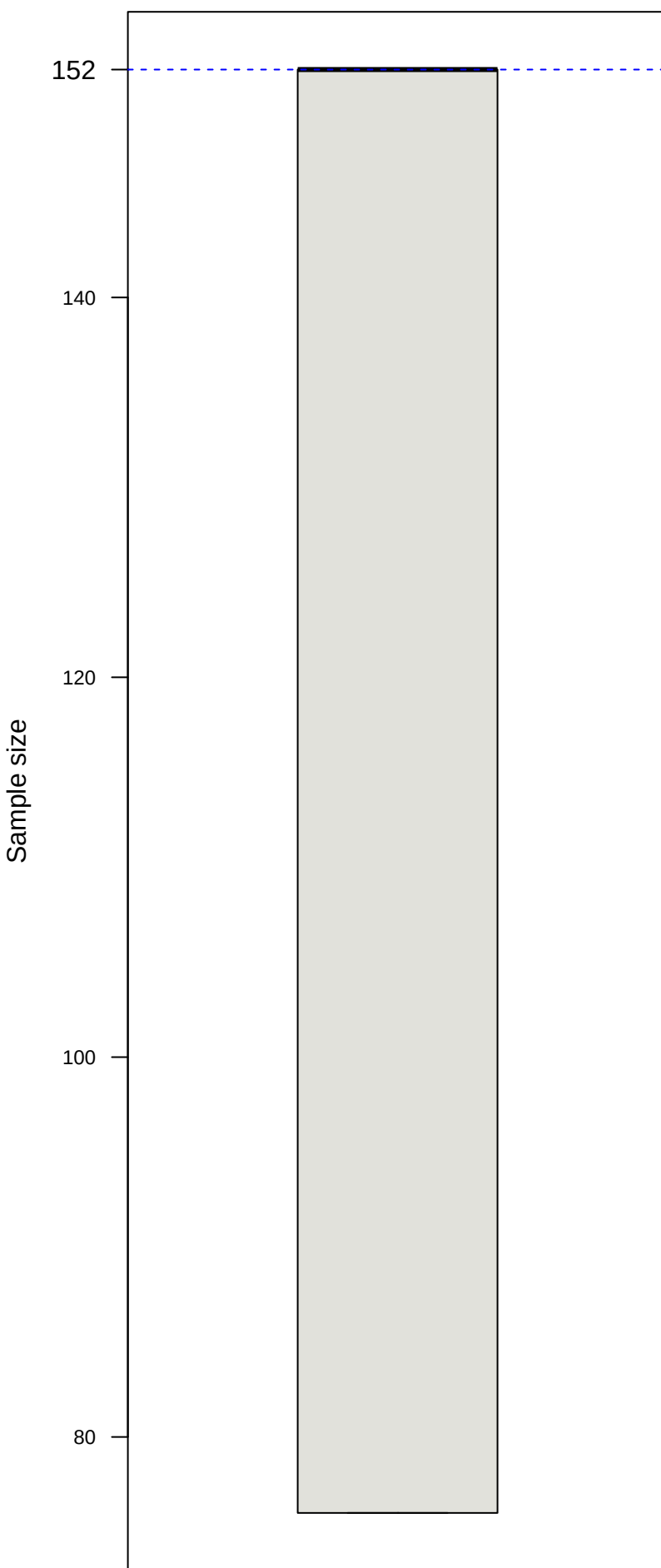
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

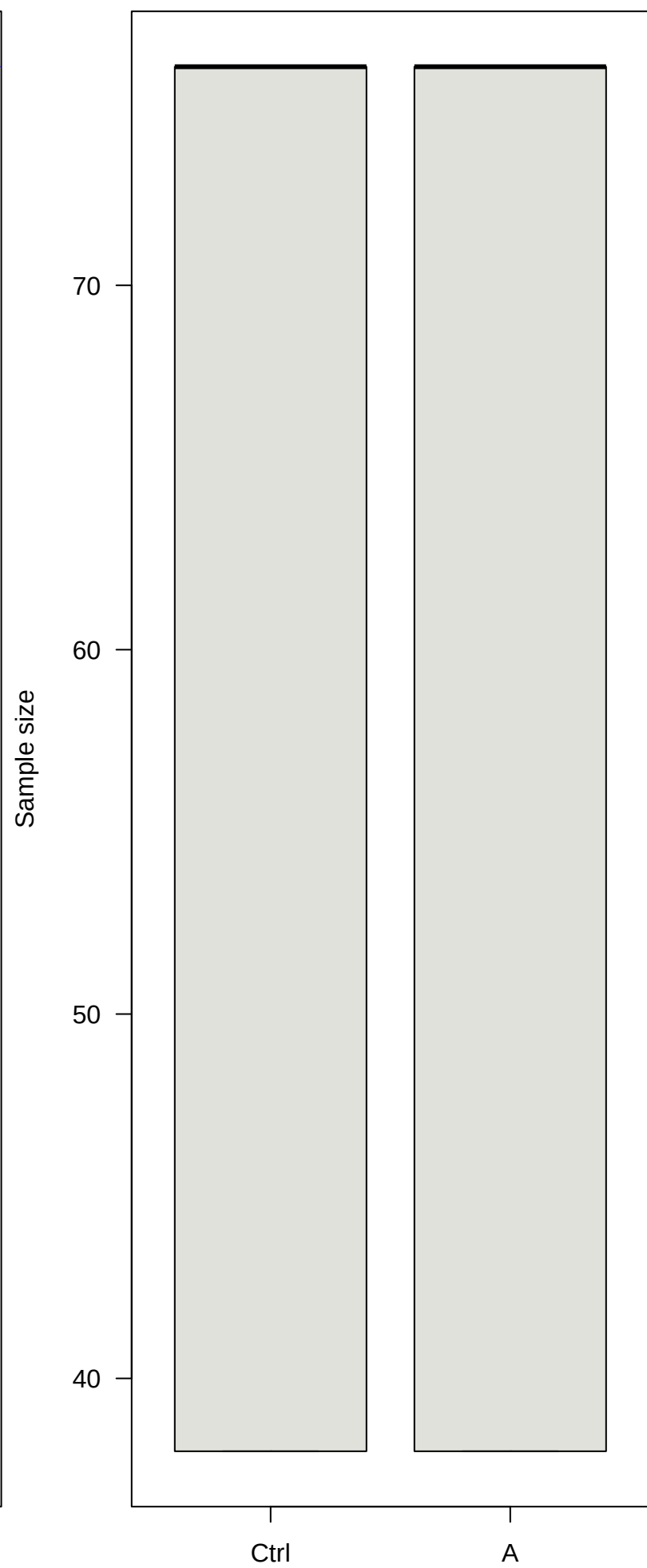
(delta/sd); residual gap from target is small-sample discreteness.

Config 118 – sample size (H1) [binary]

Total



Per group



Config 122 (0.05|0.9|1|2|0.65|0.5|0.05| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.350$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9257

FWER (P reject ≥ 1 | H0) : 0.0458

Max sample size (planned) : 64

ESS under H1 : 48.9

ESS under H0 : 48.3

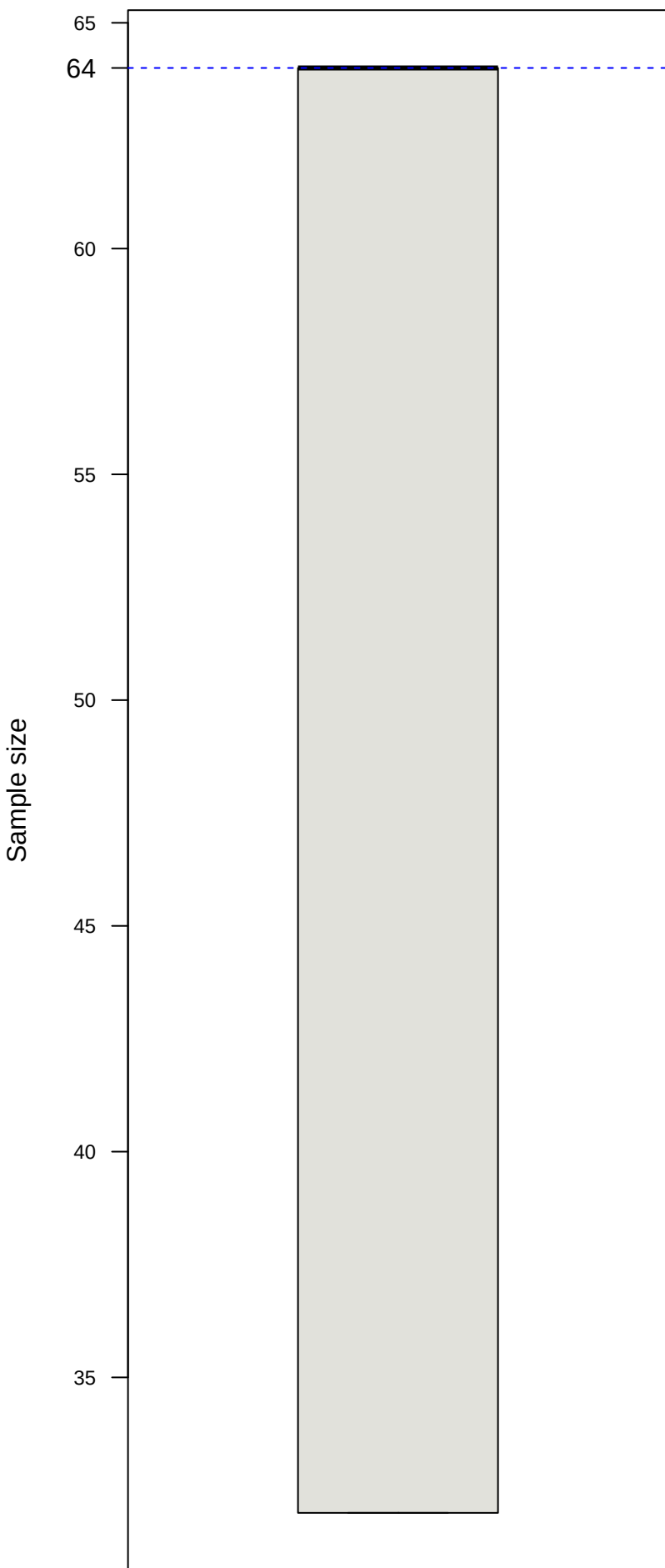
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

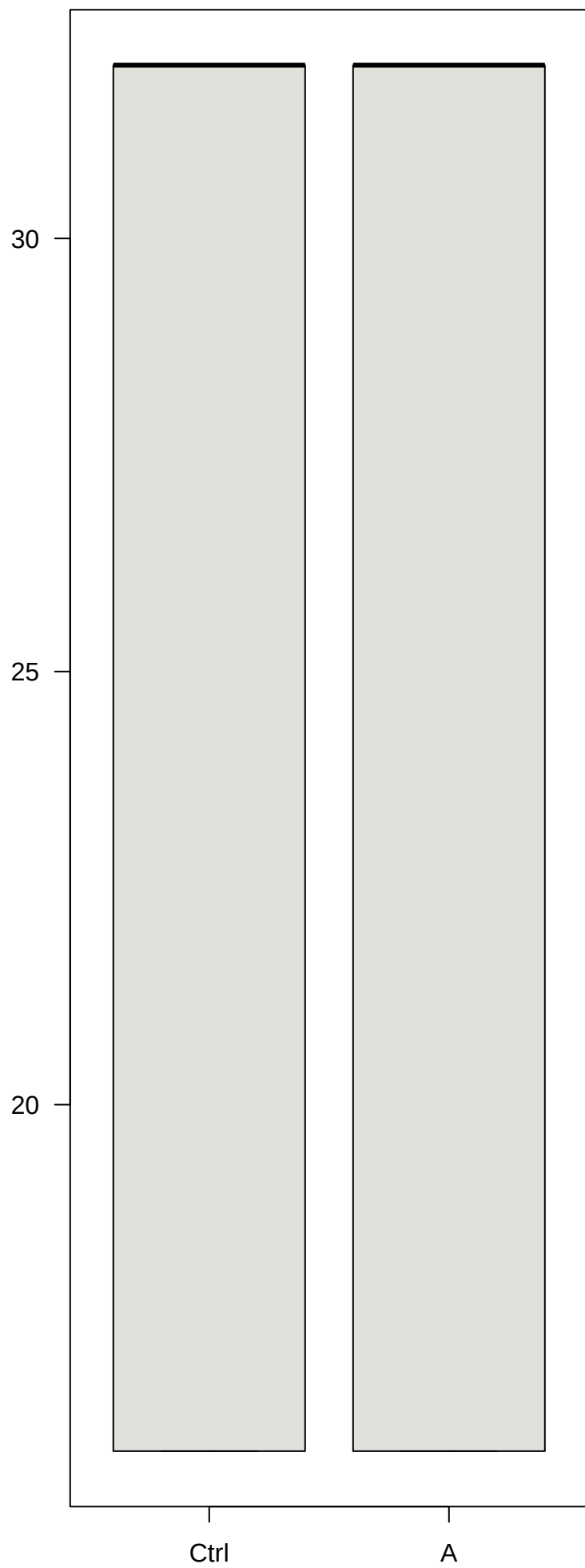
(delta/sd); residual gap from target is small-sample discreteness.

Config 122 – sample size (H1) [binary]

Total



Per group



Config 126 (0.05|0.9|1|2|0.65|0.5| 0.3| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

 effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject >=1 | H1) : 0.8956

FWER (P reject >=1 | H0) : 0.0466

Max sample size (planned) : 104

ESS under H1 : 79.0

ESS under H0 : 66.8

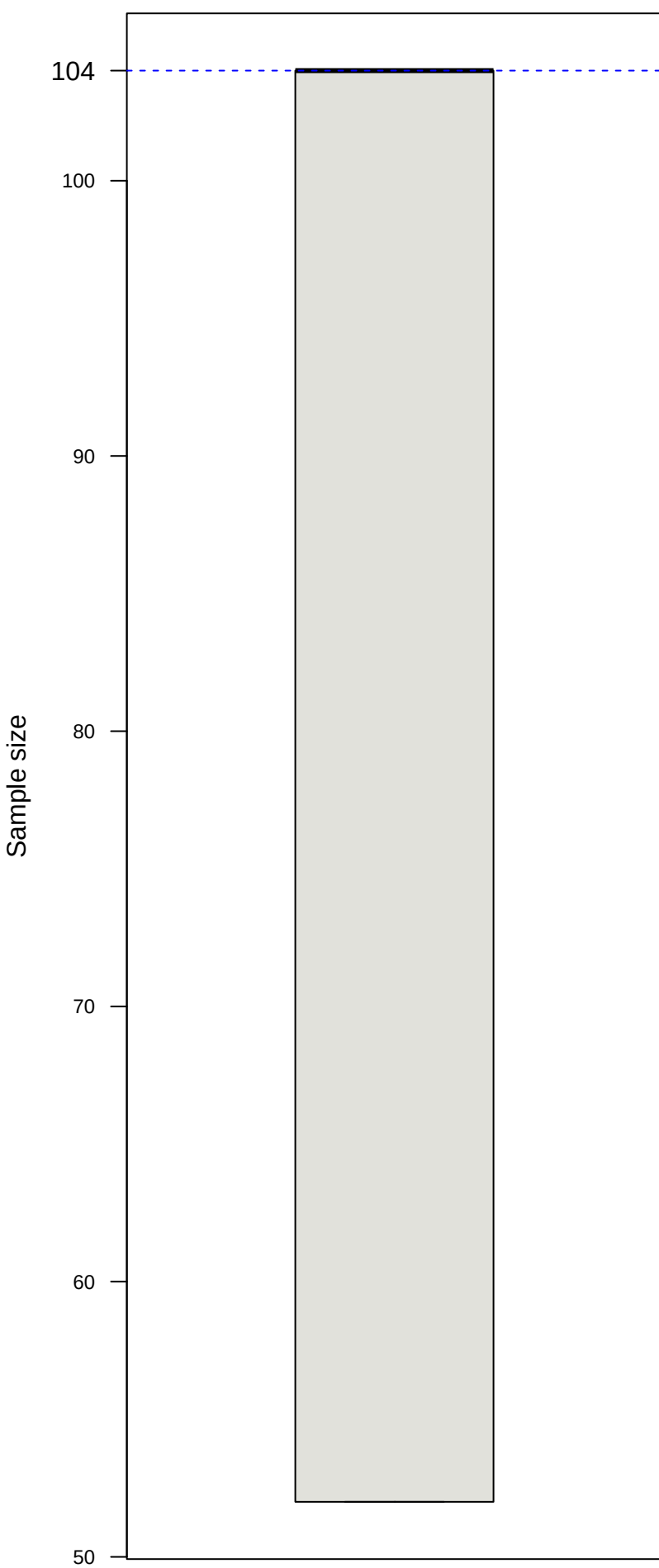
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

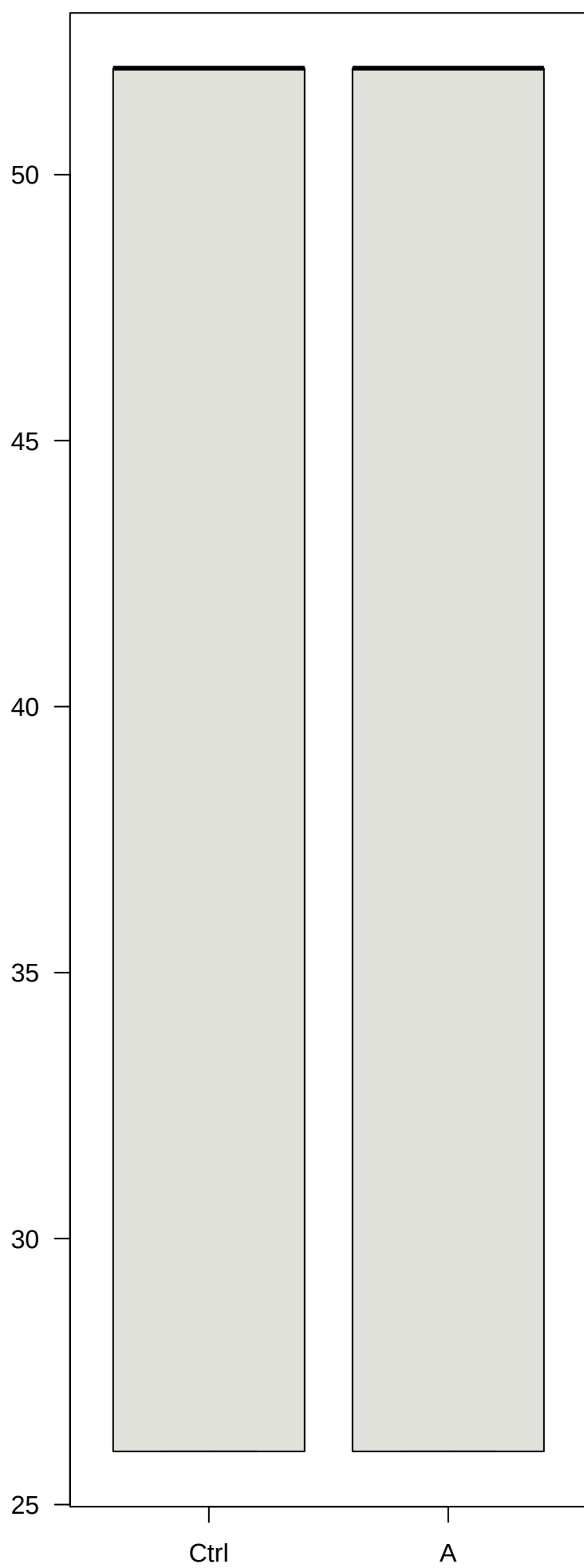
(delta/sd); residual gap from target is small-sample discreteness.

Config 126 – sample size (H1) [binary]

Total



Per group



Config 128 (0.05|0.9|1|2|0.65|0.5| 0.5| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9128

FWER (P reject ≥ 1 | H0) : 0.0381

Max sample size (planned) : 96

ESS under H1 : 72.6

ESS under H0 : 59.5

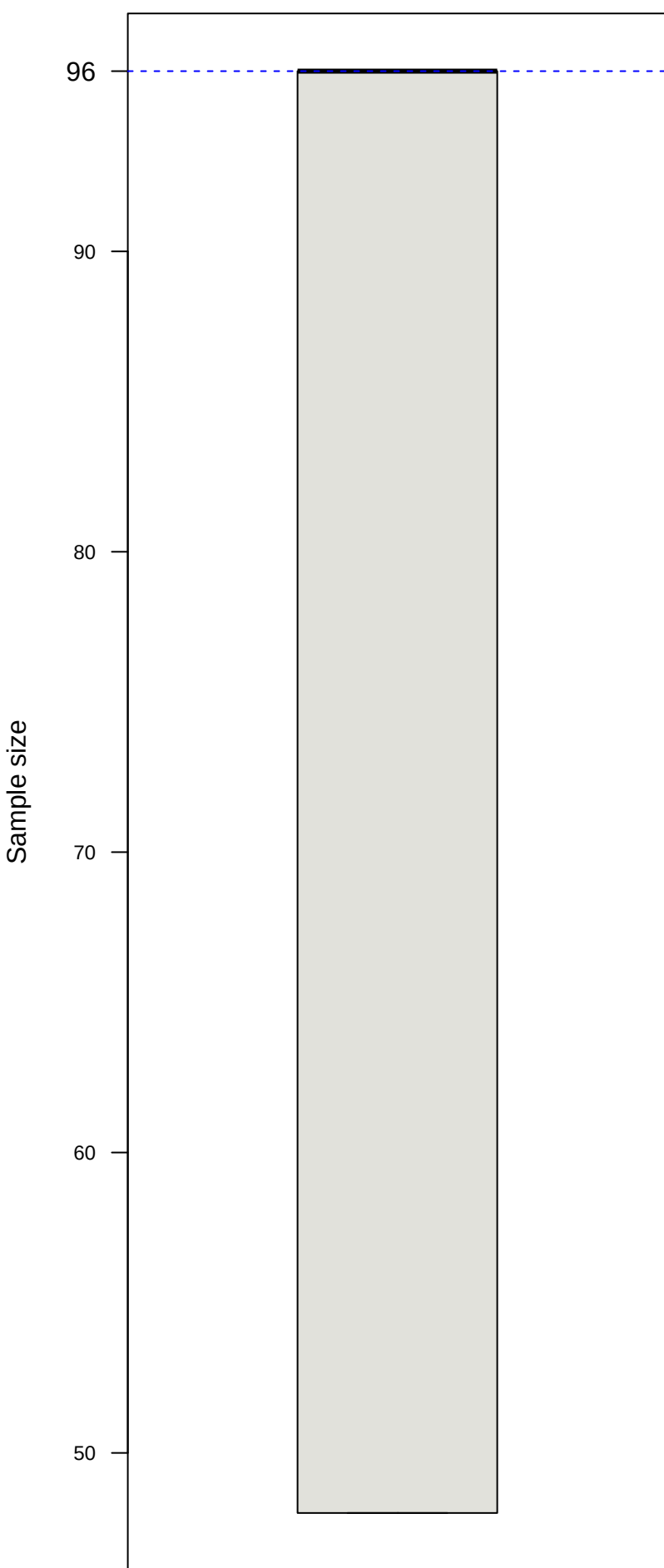
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

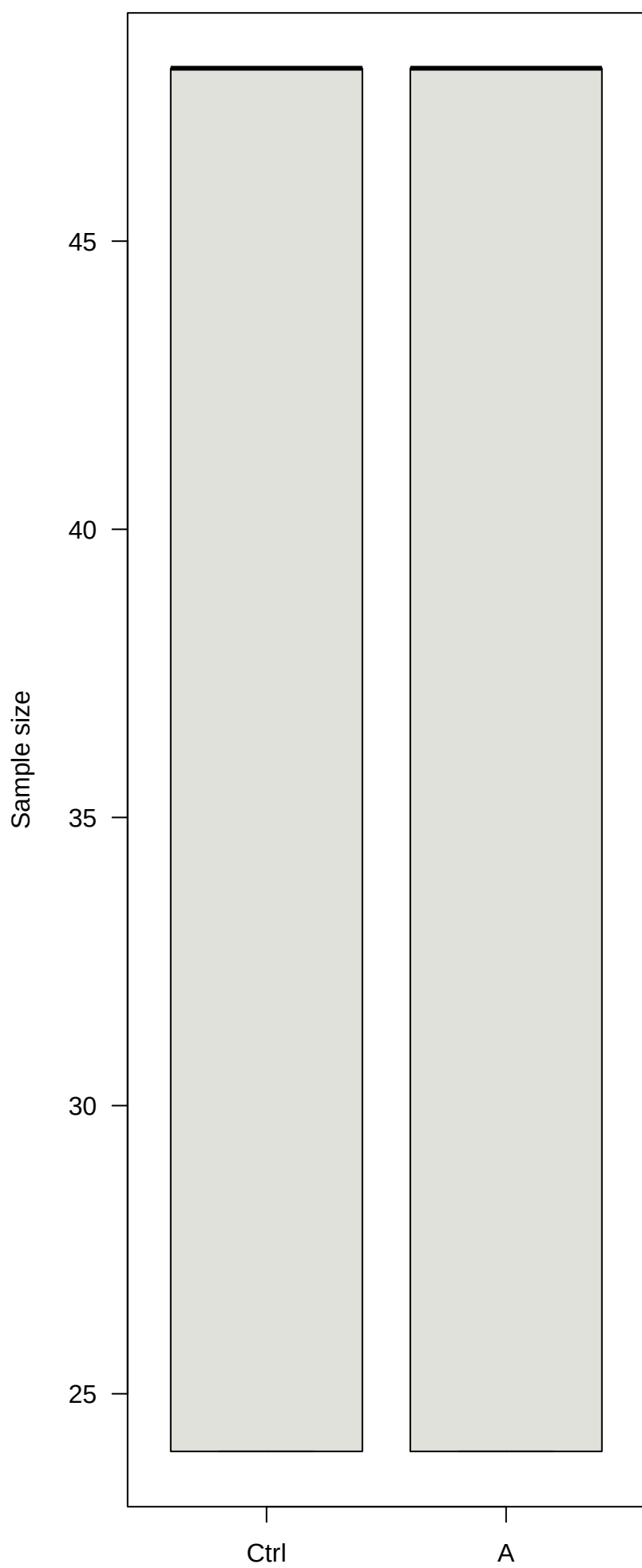
(delta/sd); residual gap from target is small-sample discreteness.

Config 128 – sample size (H1) [binary]

Total



Per group



Config 138 (0.05|0.9|1|2|0.75|0.5| 0.3| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9108

FWER (P reject ≥ 1 | H0) : 0.0474

Max sample size (planned) : 36

ESS under H1 : 27.3

ESS under H0 : 22.6

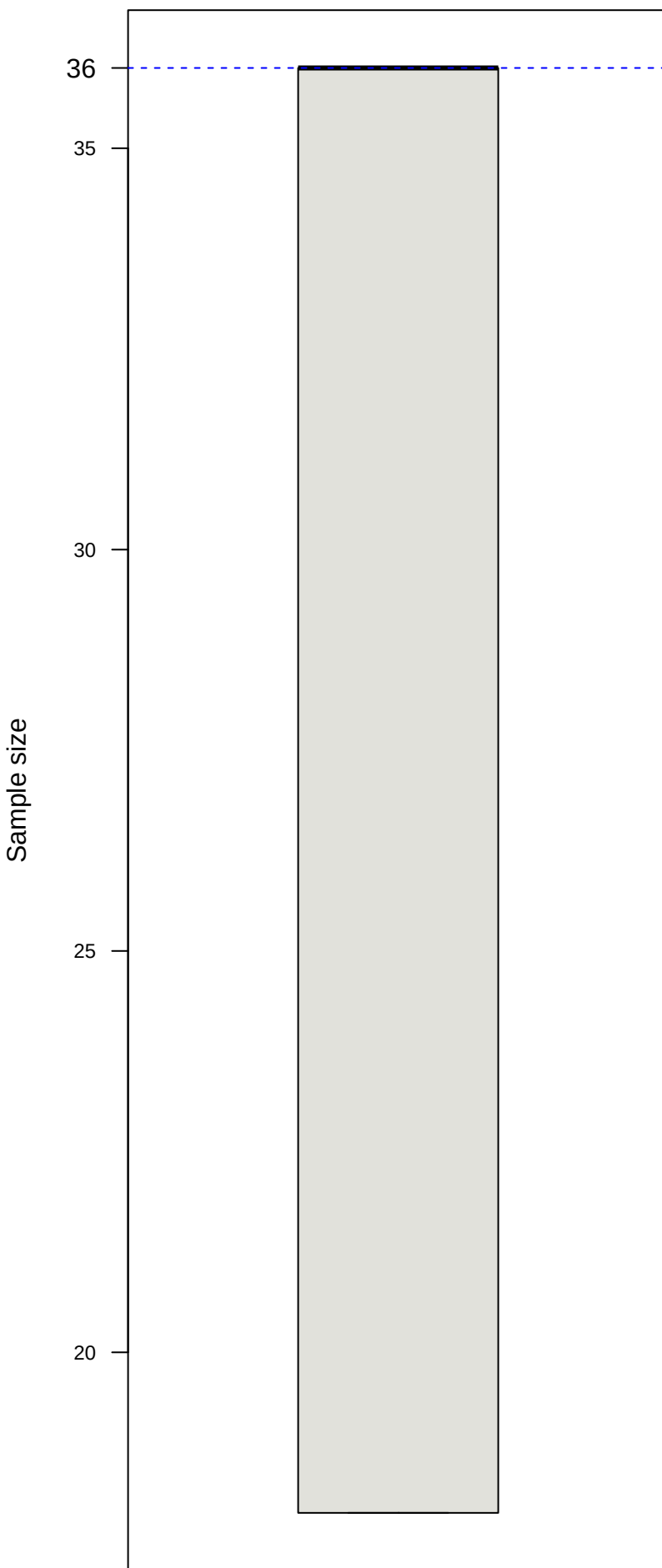
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

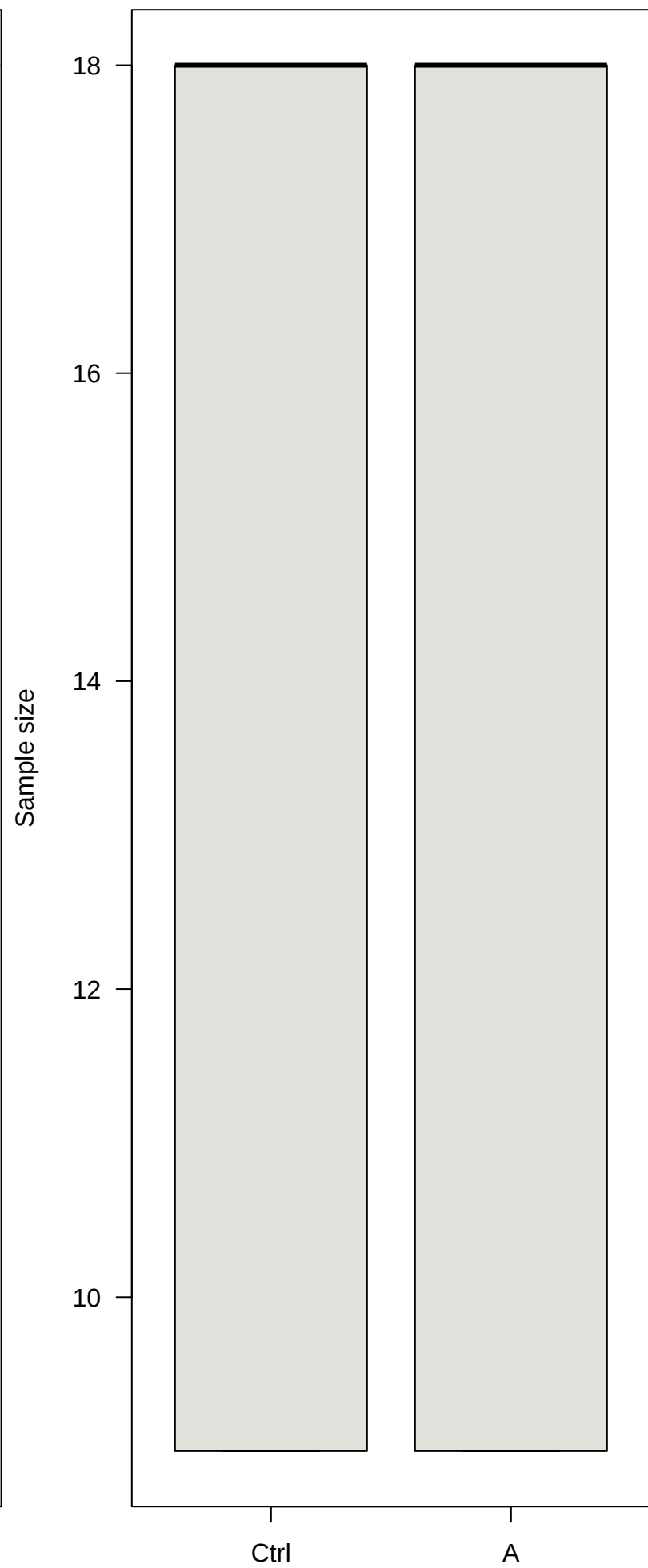
(delta/sd); residual gap from target is small-sample discreteness.

Config 138 – sample size (H1) [binary]

Total



Per group



Config 148 (0.05|0.9|1|2|0.85|0.5| 0.1| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.85

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.100

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9194

FWER (P reject ≥ 1 | H0) : 0.0170

Max sample size (planned) : 16

ESS under H1 : 13.7

ESS under H0 : 13.4

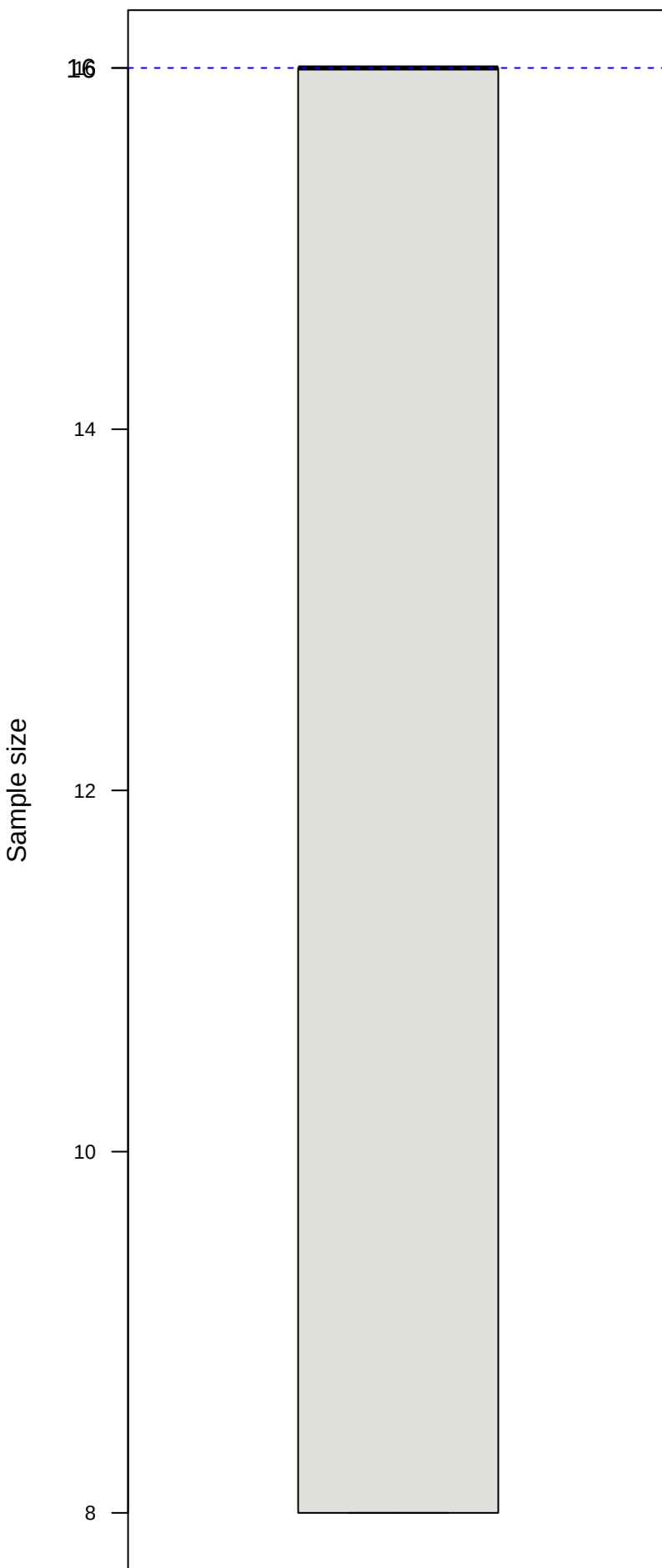
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

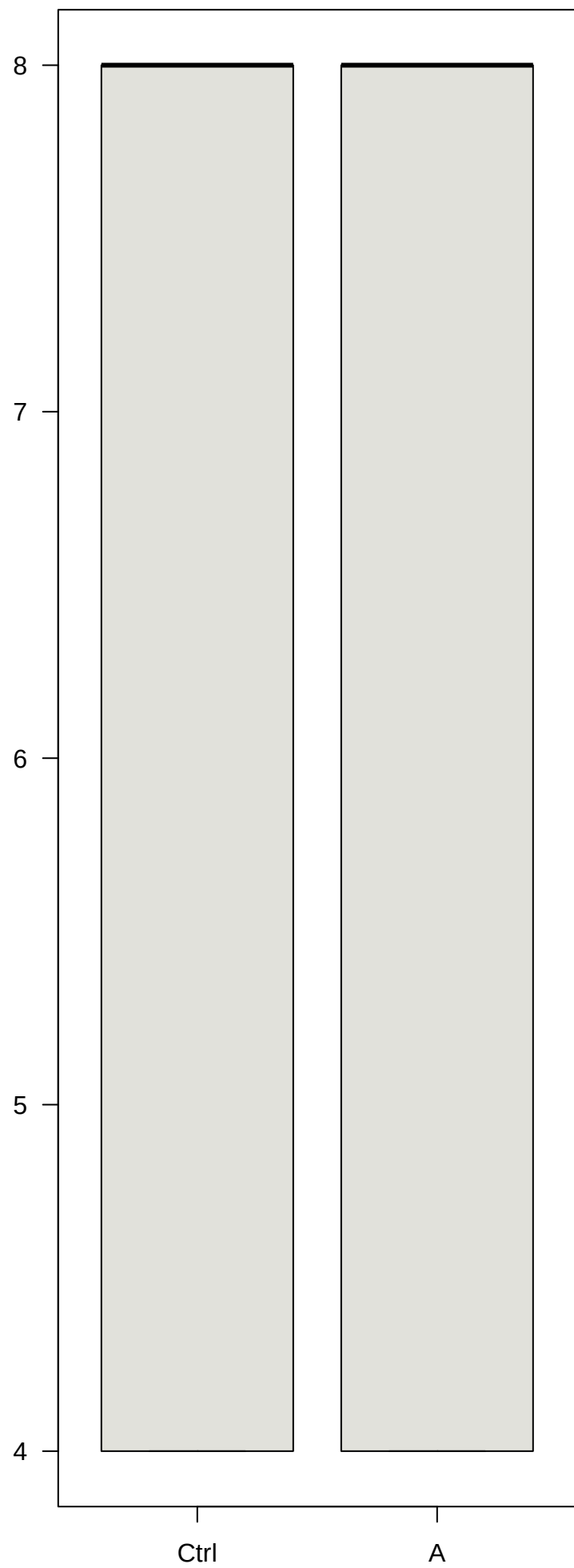
(delta/sd); residual gap from target is small-sample discreteness.

Config 148 – sample size (H1) [binary]

Total



Per group



Config 158 (0.05|0.9|1|2|0.95|0.5|0.05| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 1

J (looks) : 2

p (target win-prob) : 0.95

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.950$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9424

FWER (P reject ≥ 1 | H0) : 0.0004

Max sample size (planned) : 8

ESS under H1 : 8.0

ESS under H0 : 7.6

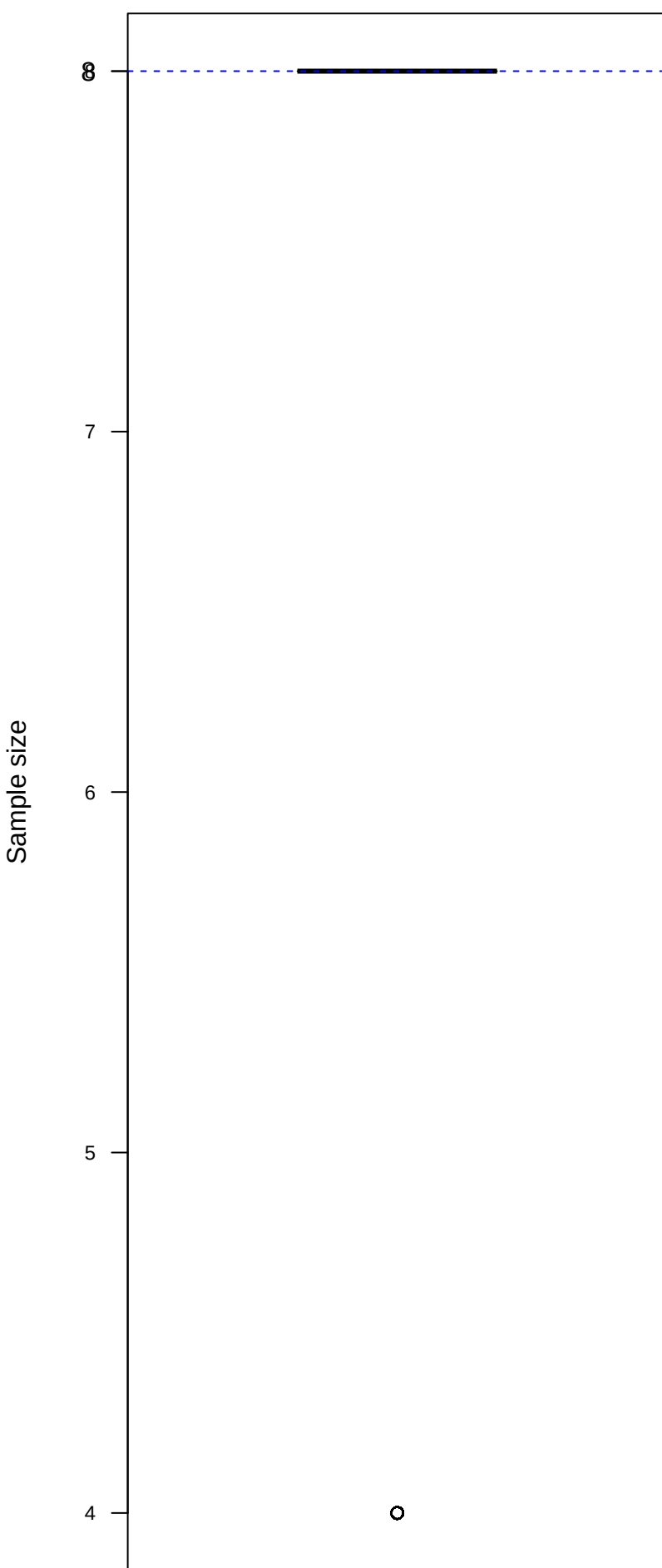
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

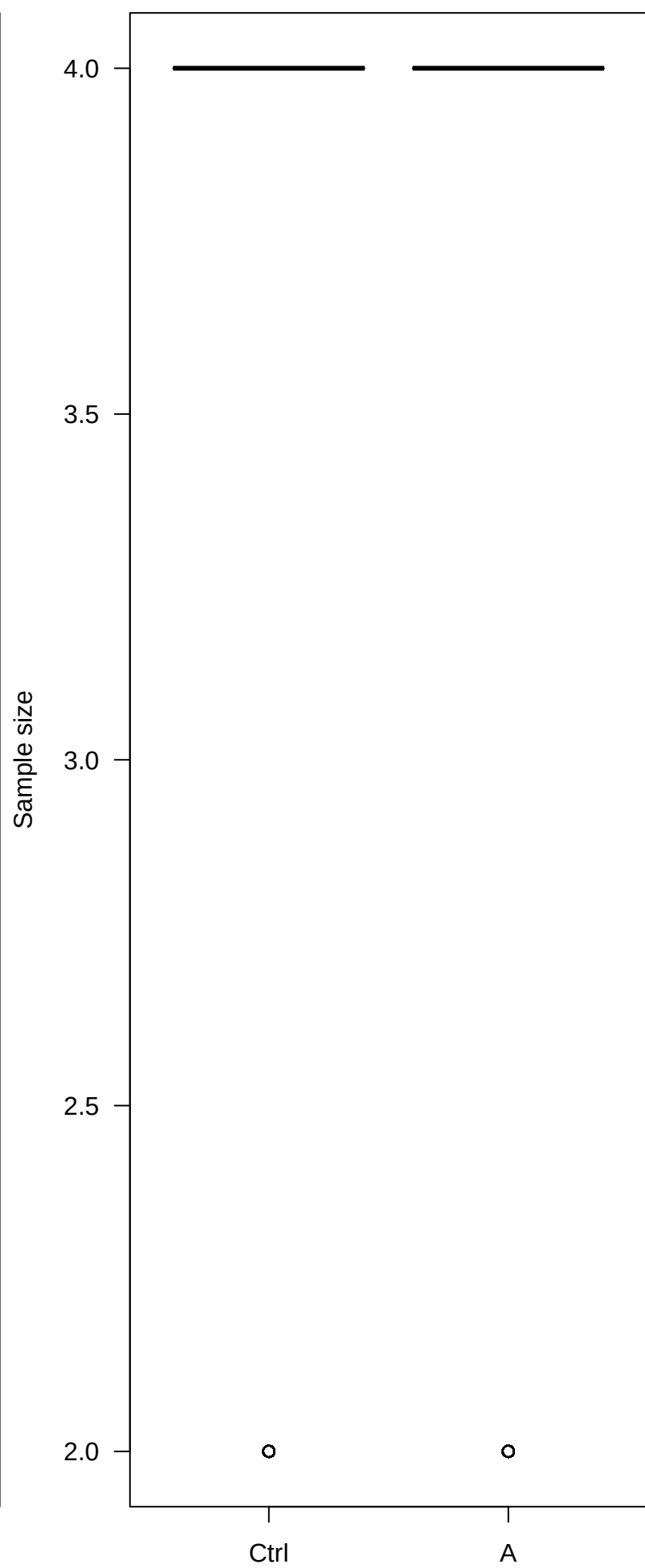
(delta/sd); residual gap from target is small-sample discreteness.

Config 158 – sample size (H1) [binary]

Total



Per group



Config 178 (0.05|0.9|4|1|0.52|0.5| 0.7| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.700$

effect $\pi_t = 0.740$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8954

FWER (P reject ≥ 1 | H0) : 0.0508

Max sample size (planned) : 14920

ESS under H1 : 14920.0

ESS under H0 : 14920.0

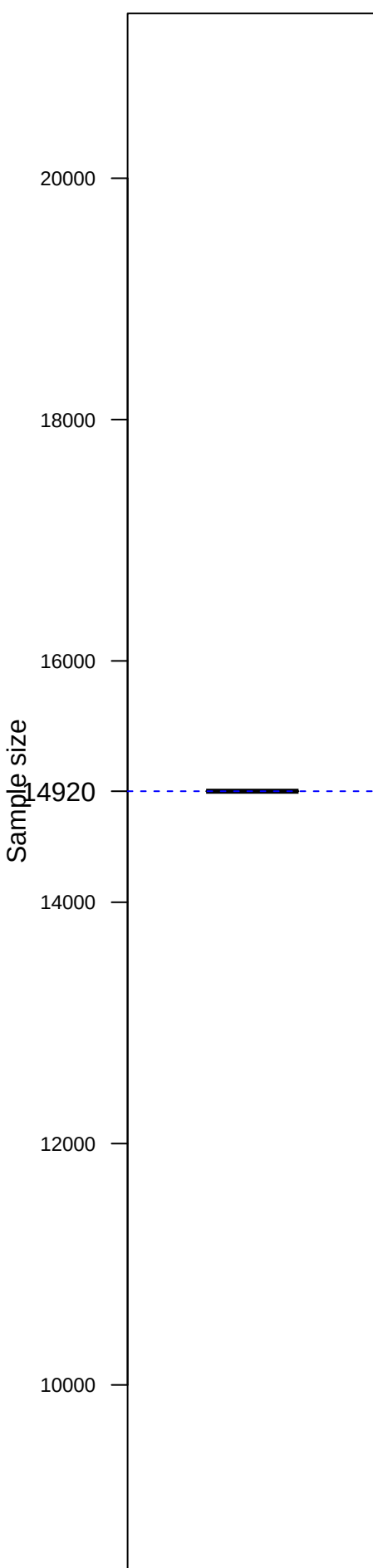
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

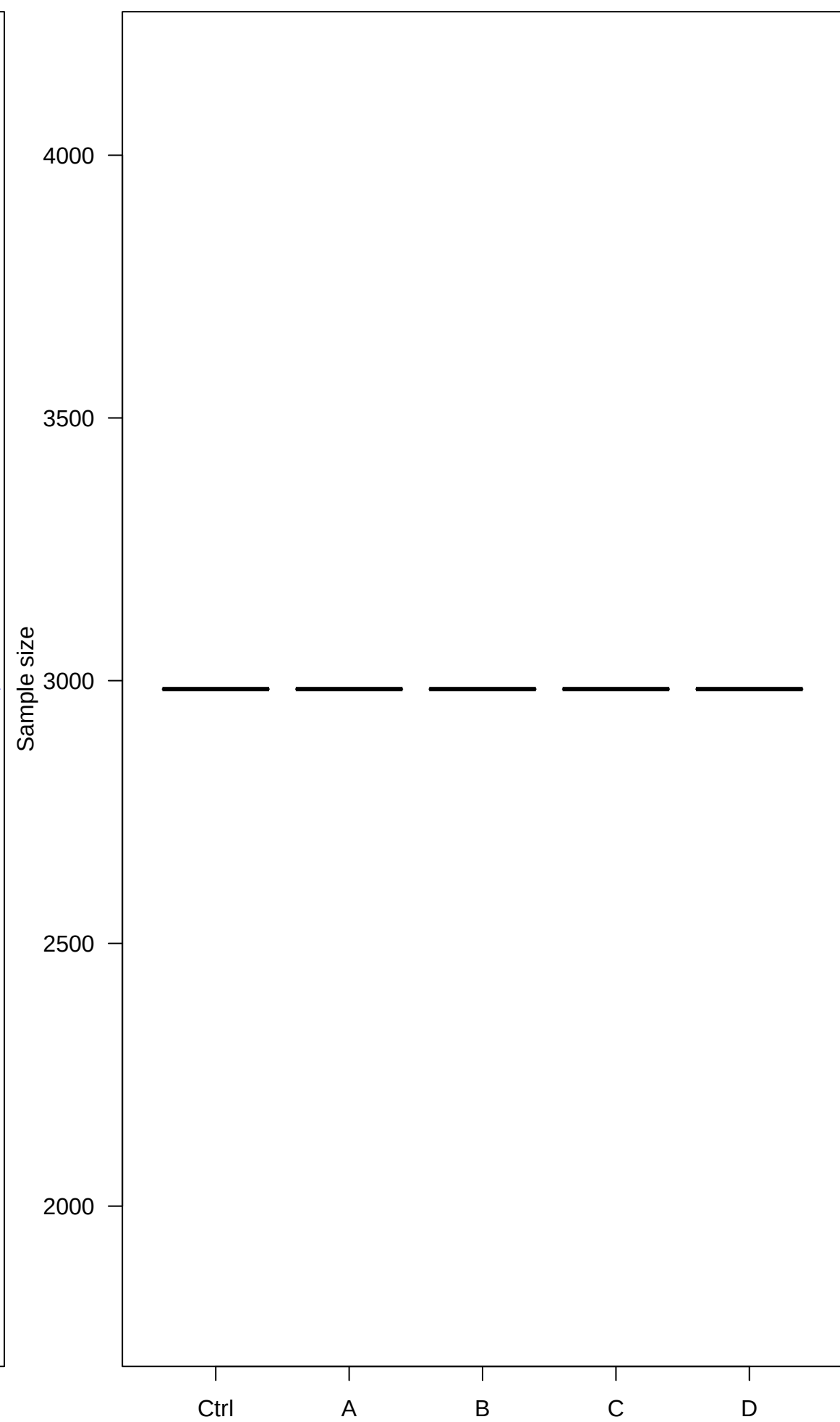
(delta/sd); residual gap from target is small-sample discreteness.

Config 178 – sample size (H1) [binary]

Total



Per group



Config 180 (0.05|0.9|4|1|0.52|0.5| 0.9| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.900

effect pi_t = 0.940 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8767

FWER (P reject ≥ 1 | H0) : 0.0496

Max sample size (planned) : 5440

ESS under H1 : 5440.0

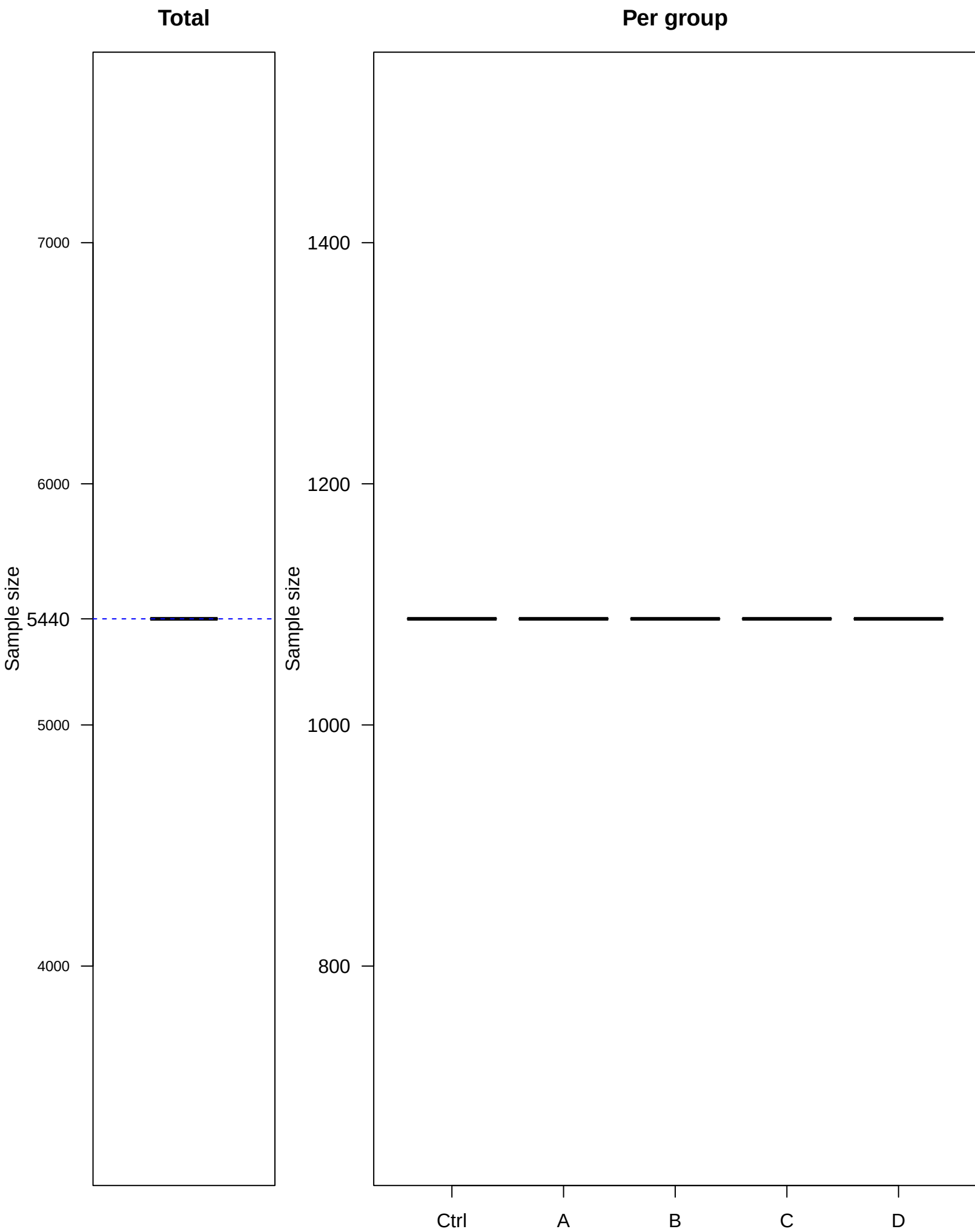
ESS under H0 : 5440.0

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 180 – sample size (H1) [binary]



Config 188 (0.05|0.9|4|1|0.55|0.5| 0.5| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.55

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9016

FWER (P reject ≥ 1 | H0) : 0.0540

Max sample size (planned) : 2925

ESS under H1 : 2925.0

ESS under H0 : 2925.0

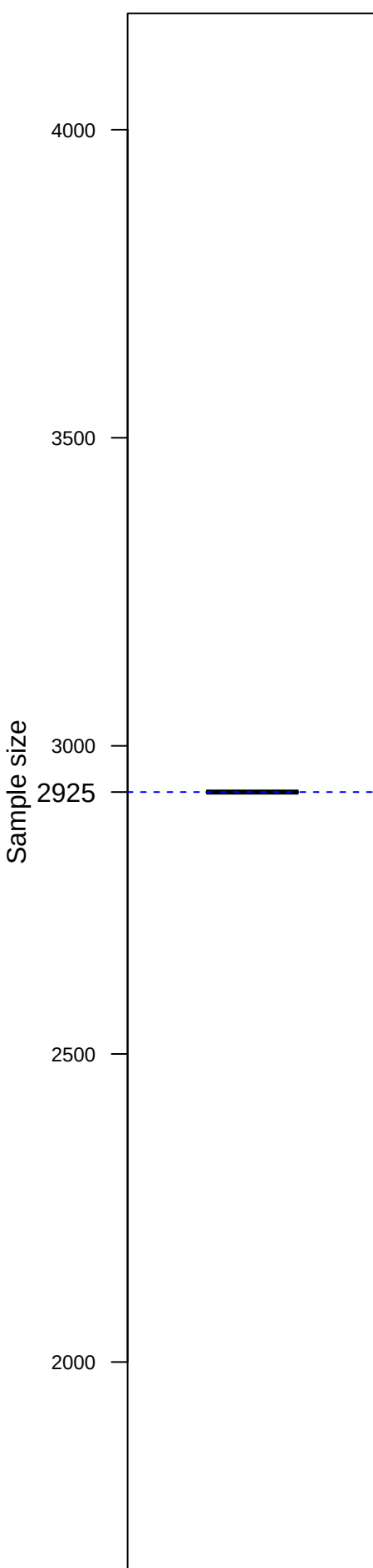
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

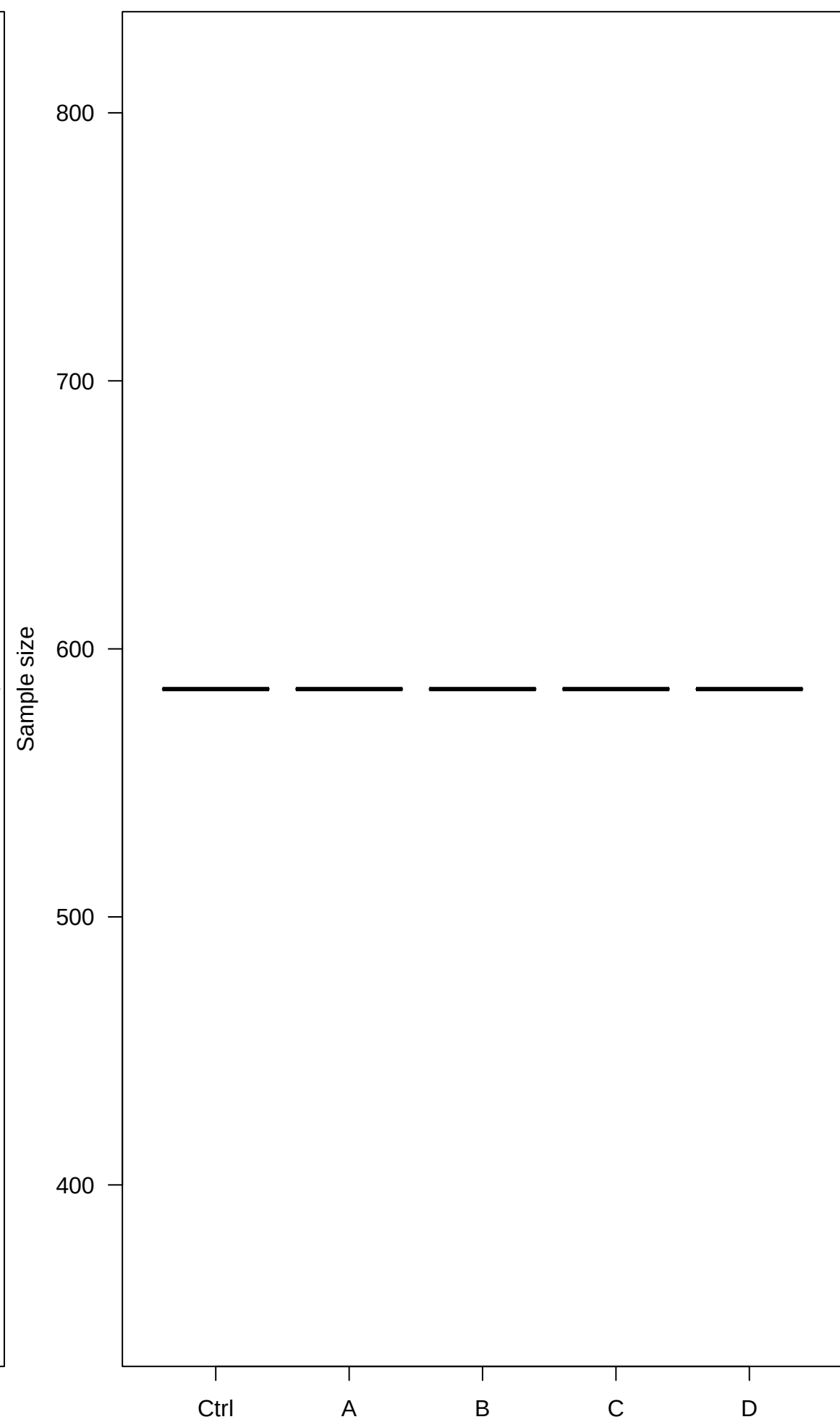
(delta/sd); residual gap from target is small-sample discreteness.

Config 188 – sample size (H1) [binary]

Total



Per group



Config 202 (0.05|0.9|4|1| 0.6|0.5| 0.7| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.6

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.700

effect pi_t = 0.900 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8600

FWER (P reject ≥ 1 | H0) : 0.0478

Max sample size (planned) : 465

ESS under H1 : 465.0

ESS under H0 : 465.0

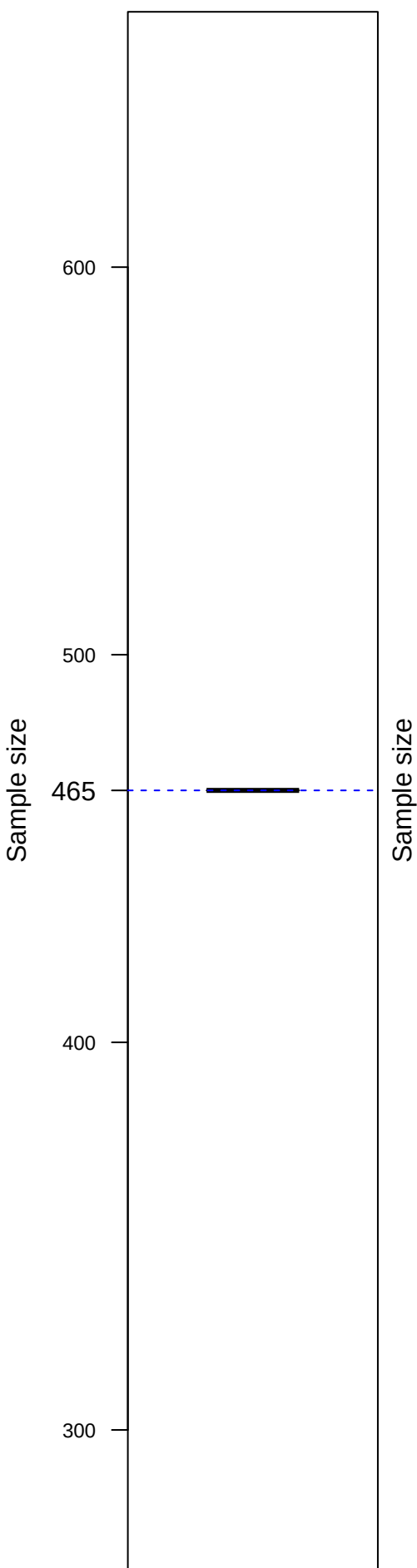
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

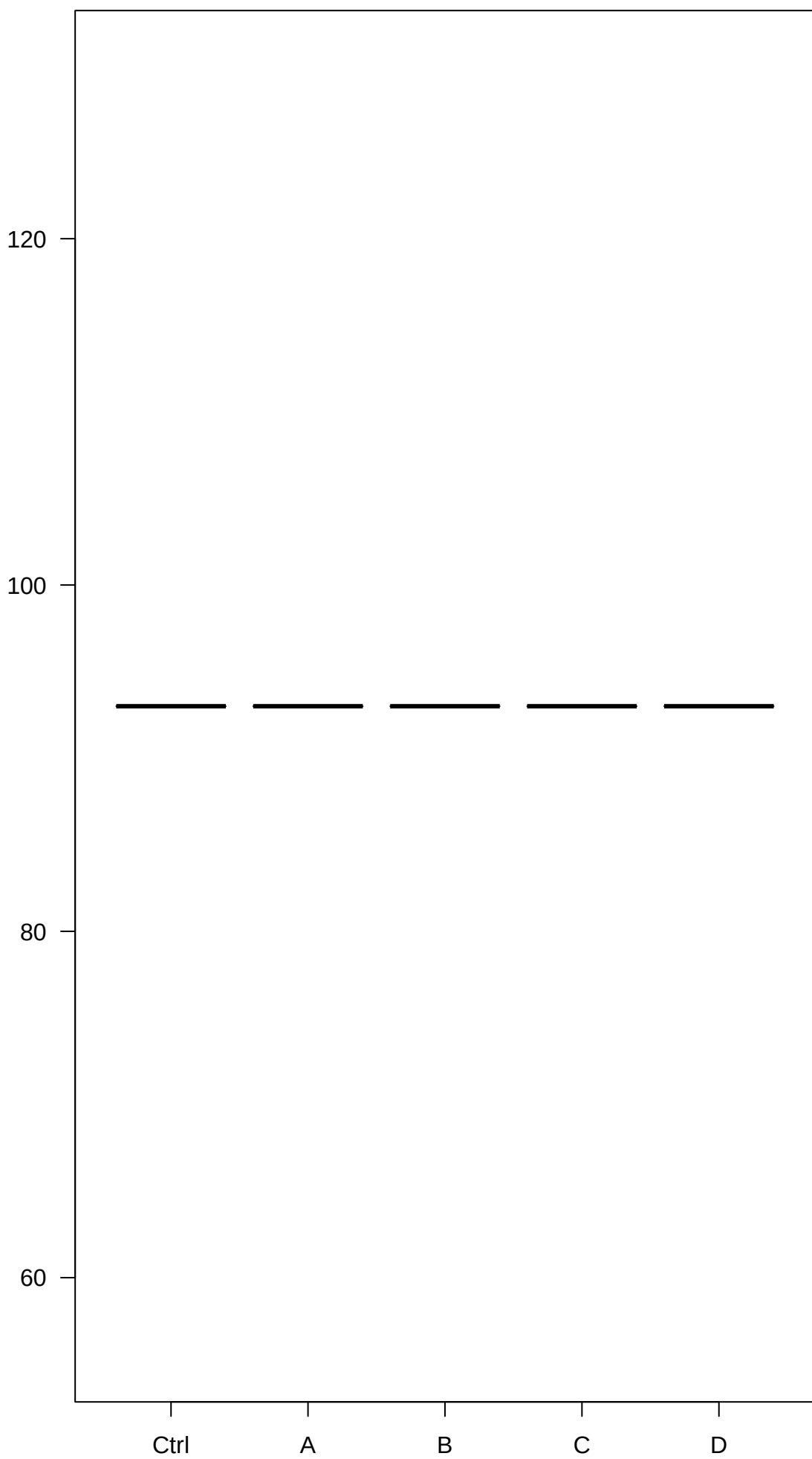
(delta/sd); residual gap from target is small-sample discreteness.

Config 202 – sample size (H1) [binary]

Total



Per group



Config 206 (0.05|0.9|4|1|0.65|0.5|0.05| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.350$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9747

FWER (P reject ≥ 1 | H0) : 0.0528

Max sample size (planned) : 200

ESS under H1 : 200.0

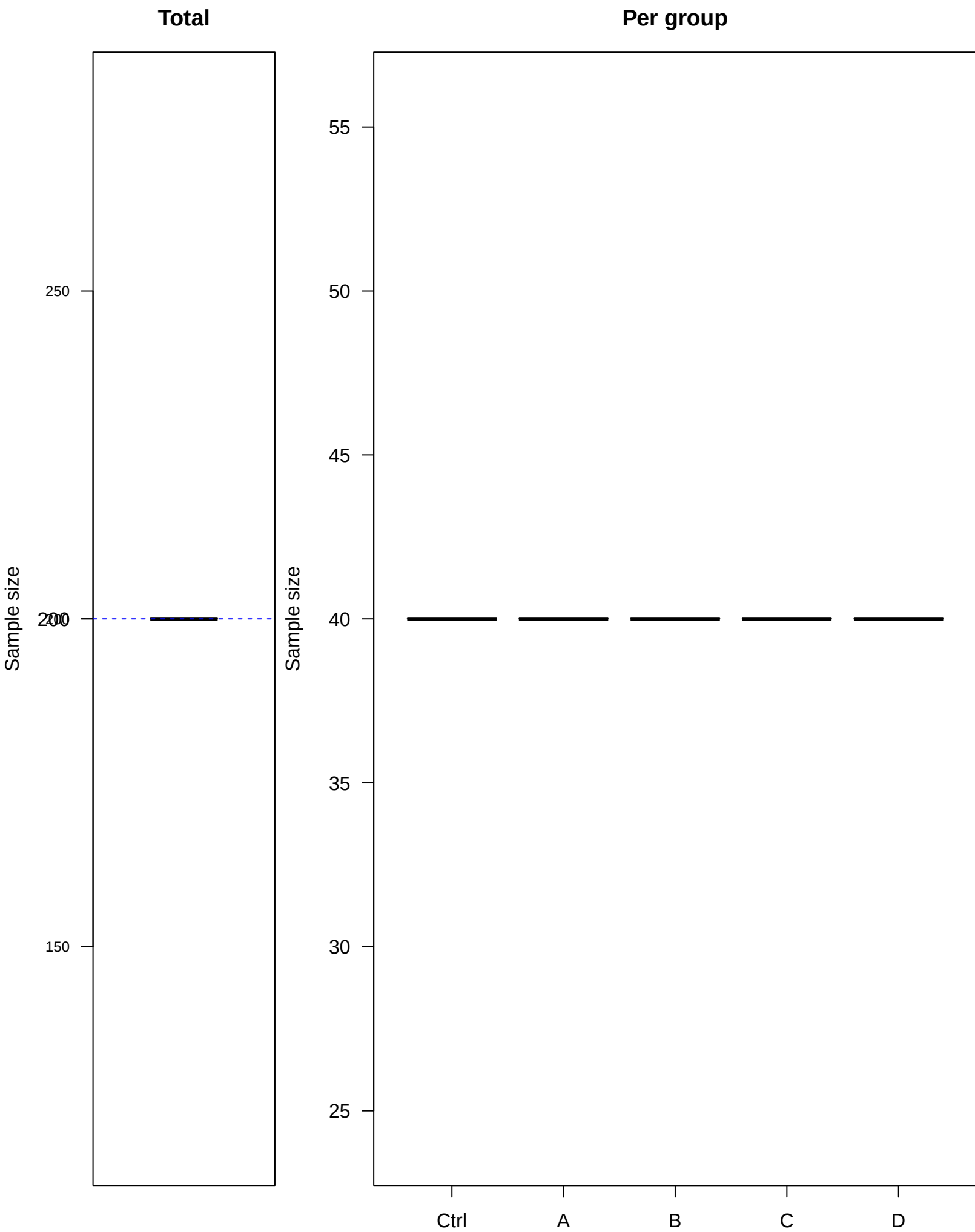
ESS under H0 : 200.0

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 206 – sample size (H1) [binary]



Config 210 (0.05|0.9|4|1|0.65|0.5| 0.3| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9155

FWER (P reject ≥ 1 | H0) : 0.0496

Max sample size (planned) : 315

ESS under H1 : 315.0

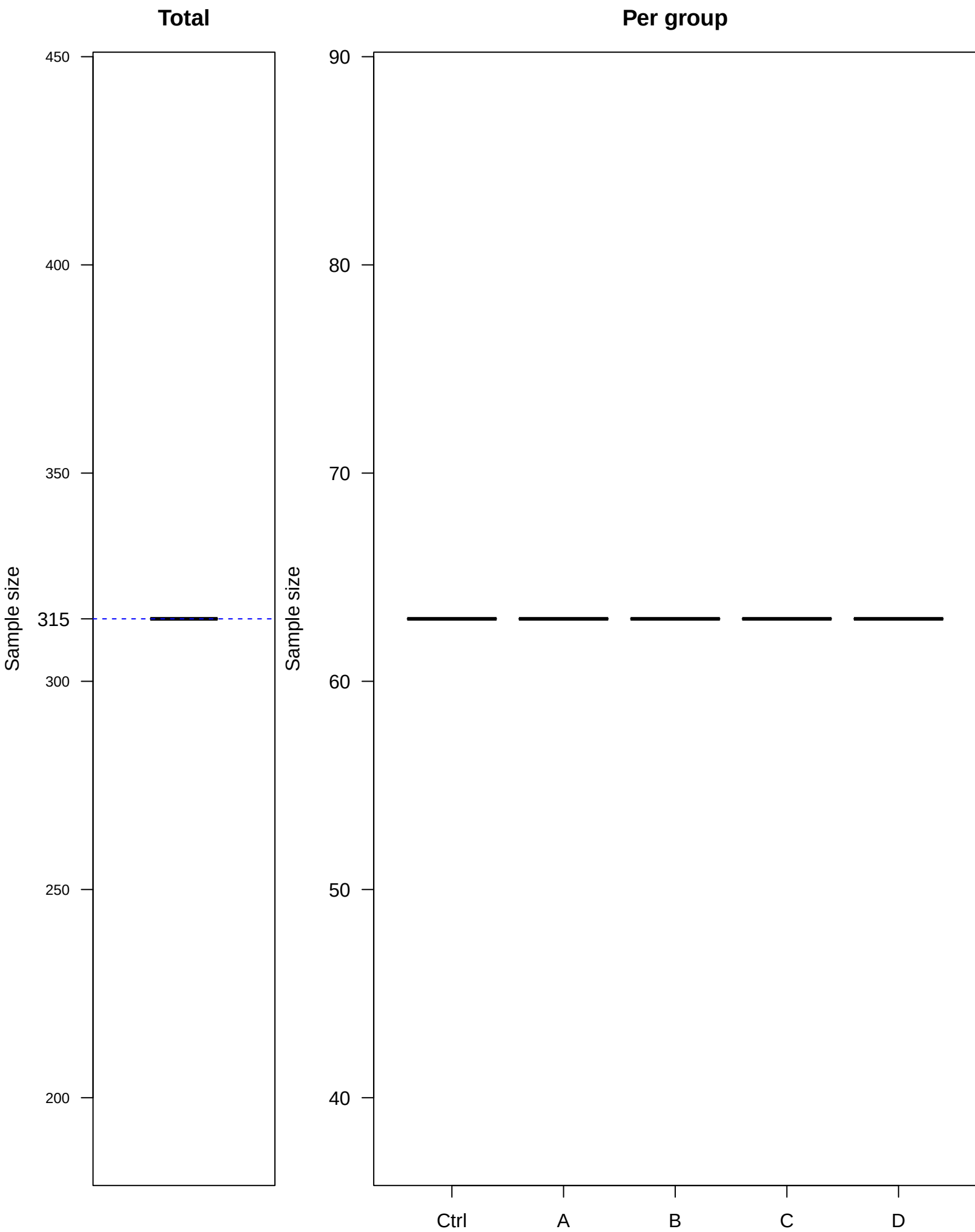
ESS under H0 : 315.0

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 210 – sample size (H1) [binary]



Config 212 (0.05|0.9|4|1|0.65|0.5| 0.5| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8868

FWER (P reject ≥ 1 | H0) : 0.0509

Max sample size (planned) : 290

ESS under H1 : 290.0

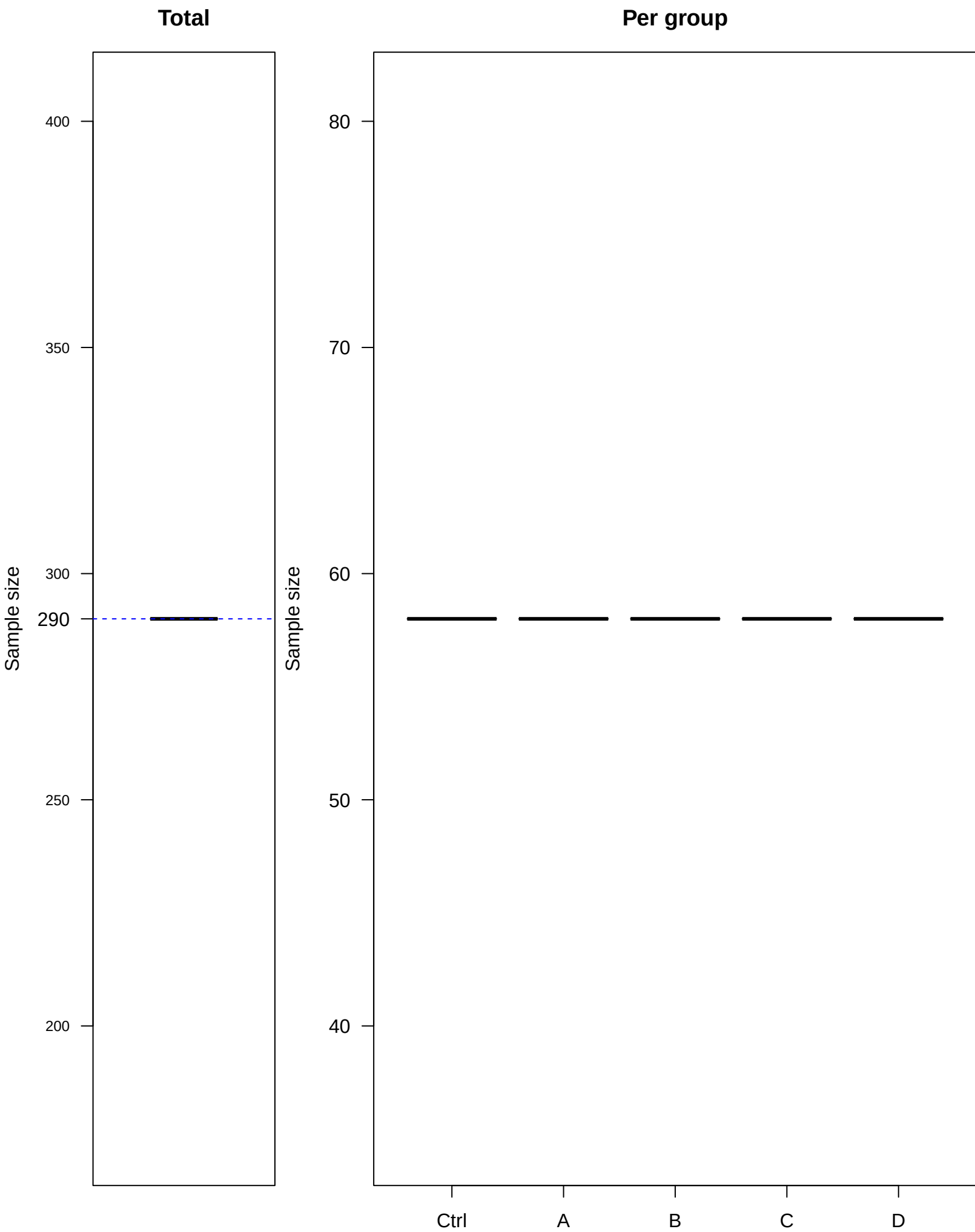
ESS under H0 : 290.0

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 212 – sample size (H1) [binary]



Config 222 (0.05|0.9|4|1|0.75|0.5| 0.3| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9054

FWER (P reject ≥ 1 | H0) : 0.0562

Max sample size (planned) : 110

ESS under H1 : 110.0

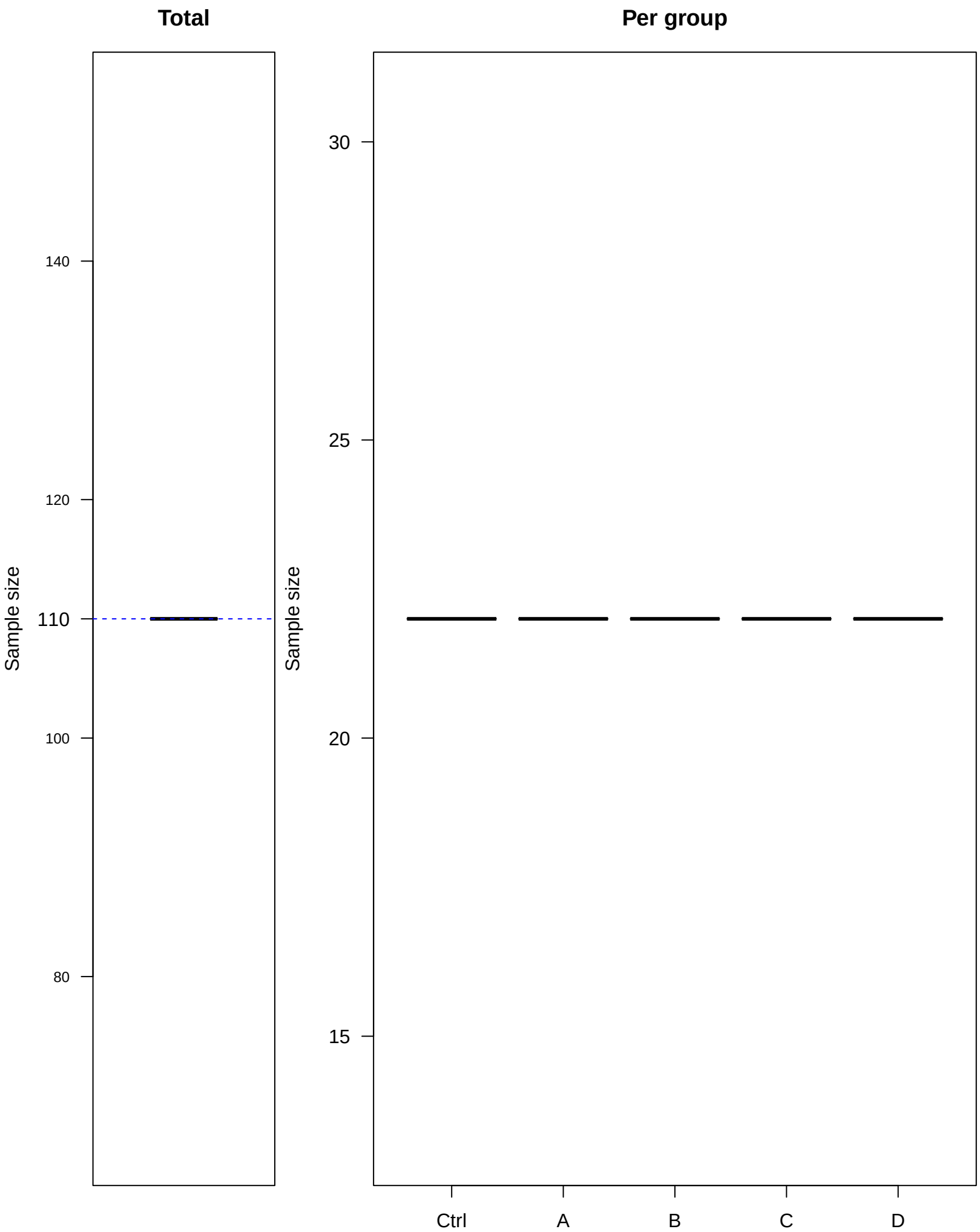
ESS under H0 : 110.0

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 222 – sample size (H1) [binary]



Config 232 (0.05|0.9|4|1|0.85|0.5| 0.1| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.85

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.100$

effect $\pi_t = 0.800$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9524

FWER (P reject ≥ 1 | H0) : 0.0652

Max sample size (planned) : 50

ESS under H1 : 50.0

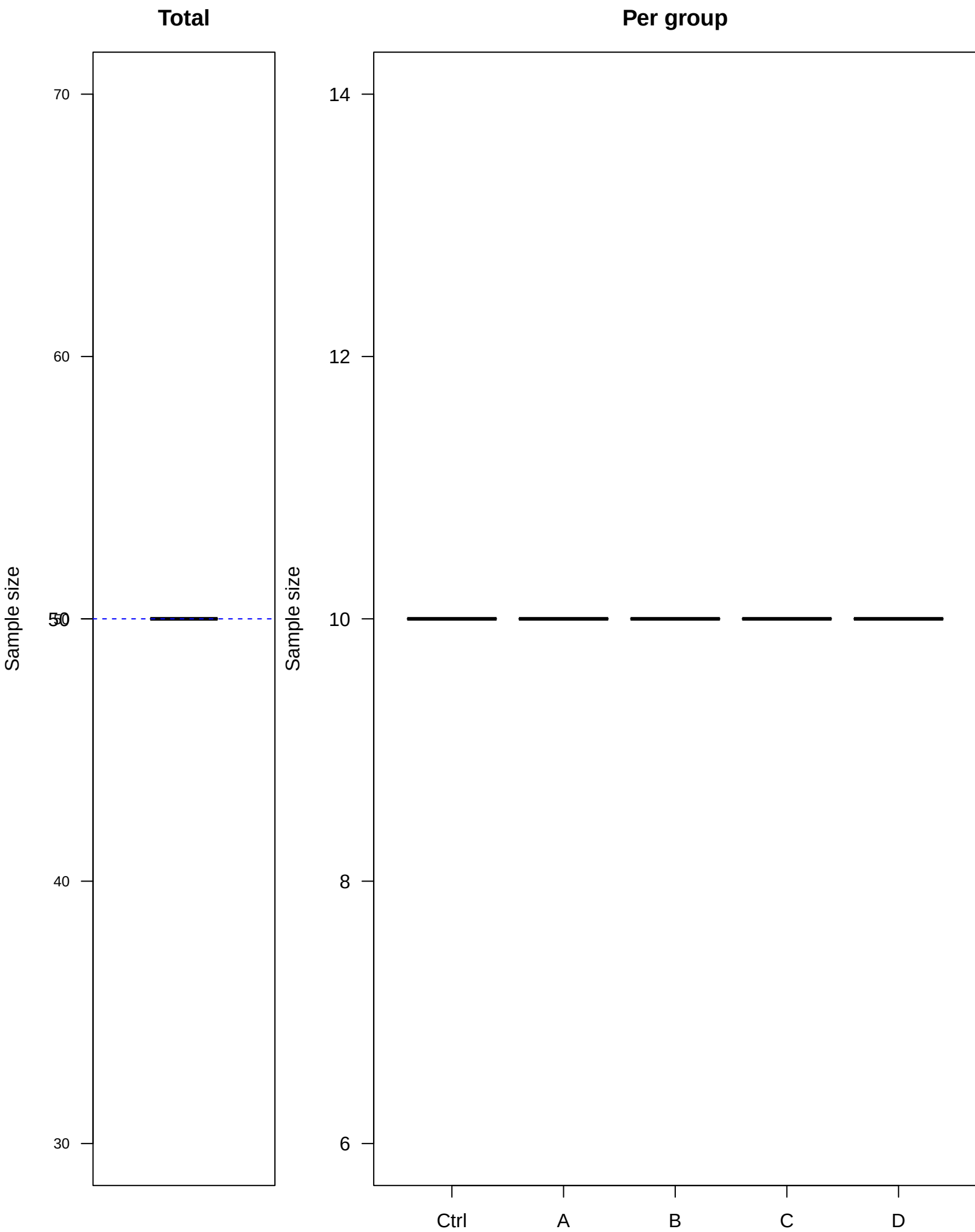
ESS under H0 : 50.0

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 232 – sample size (H1) [binary]



Config 242 (0.05|0.9|4|1|0.95|0.5|0.05| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 1

p (target win-prob) : 0.95

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.950$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9643

FWER (P reject ≥ 1 | H0) : 0.0345

Max sample size (planned) : 25

ESS under H1 : 25.0

ESS under H0 : 25.0

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 242 – sample size (H1) [binary]



Config 262 (0.05|0.9|4|2|0.52|0.5| 0.7| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.700

effect pi_t = 0.740 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9012

FWER (P reject ≥ 1 | H0) : 0.0482

Max sample size (planned) : 15930

ESS under H1 : 11532.9

ESS under H0 : 10158.0

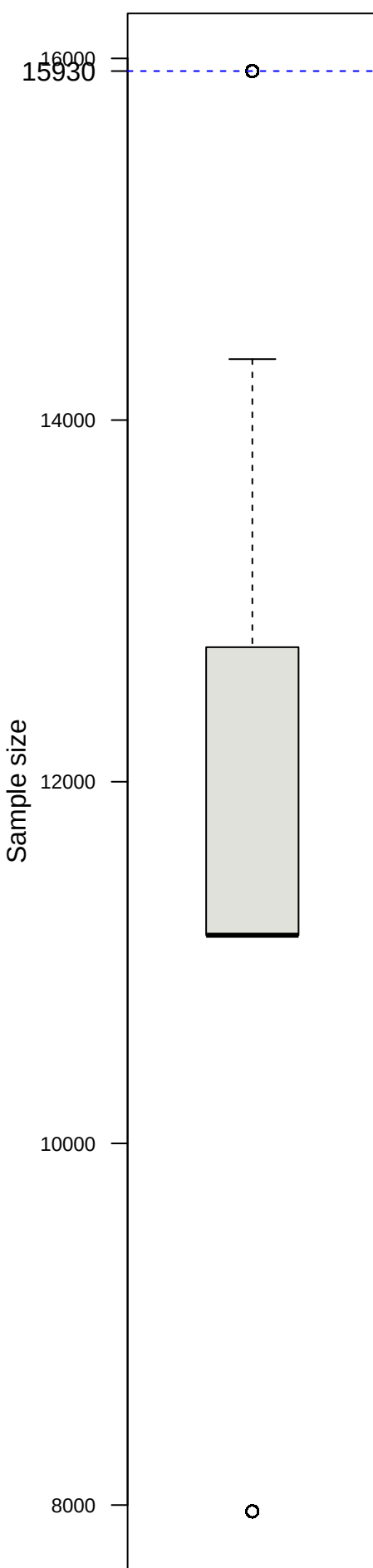
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

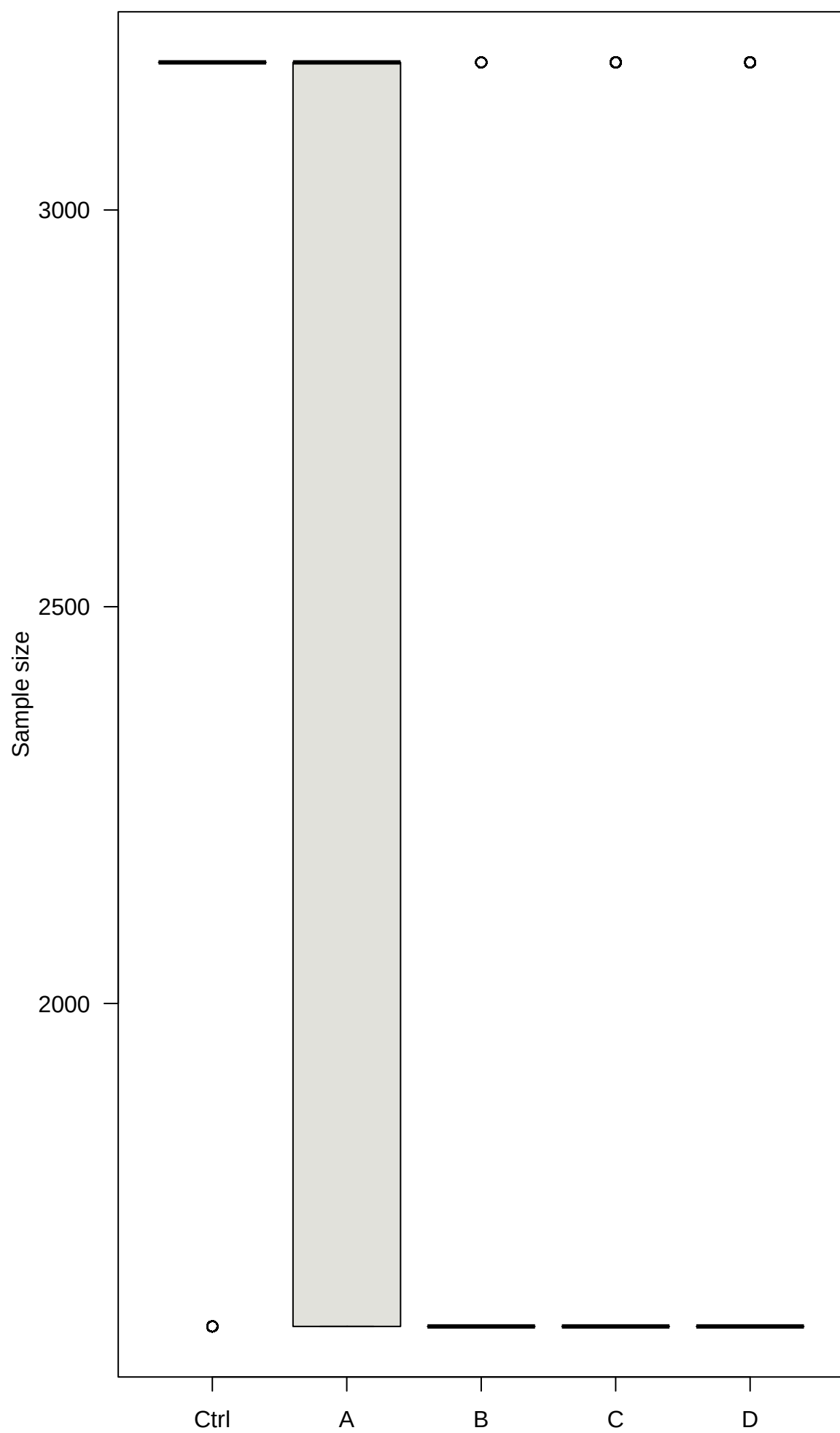
(delta/sd); residual gap from target is small-sample discreteness.

Config 262 – sample size (H1) [binary]

Total



Per group



Config 264 (0.05|0.9|4|2|0.52|0.5| 0.9| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.52

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.900

effect pi_t = 0.940 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8850

FWER (P reject ≥ 1 | H0) : 0.0452

Max sample size (planned) : 5810

ESS under H1 : 4265.4

ESS under H0 : 3709.0

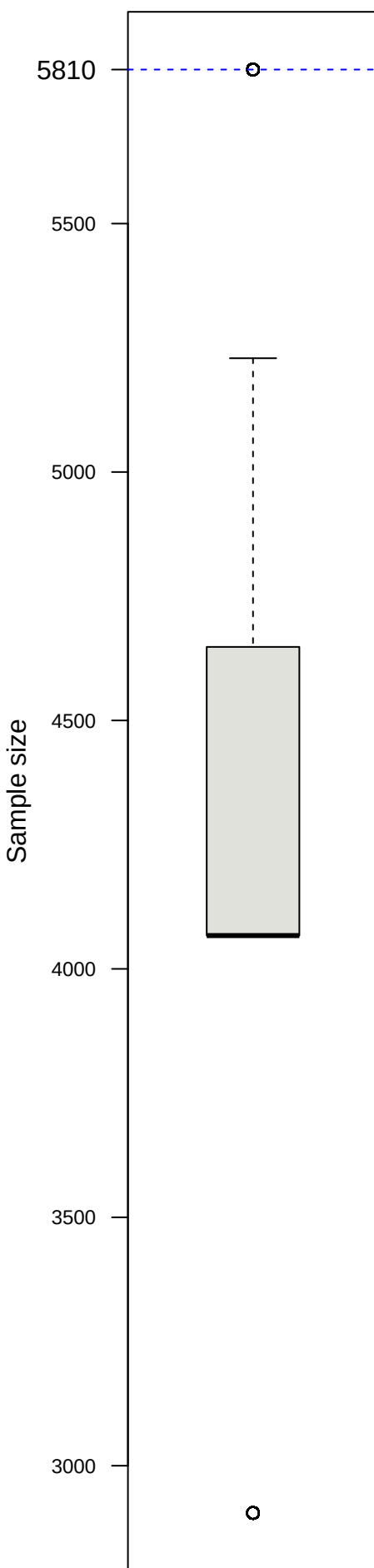
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

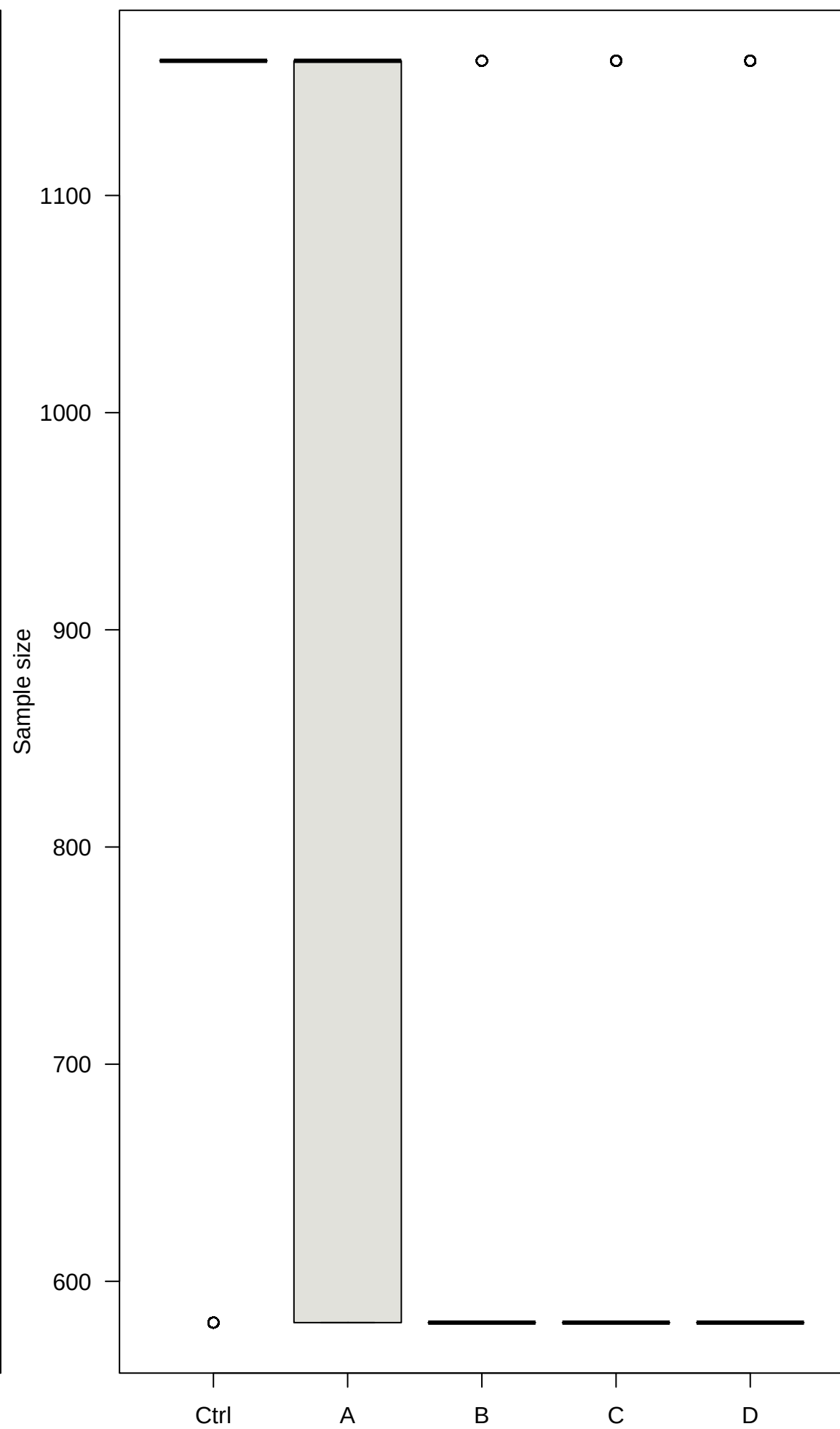
(delta/sd); residual gap from target is small-sample discreteness.

Config 264 – sample size (H1) [binary]

Total



Per group



Config 272 (0.05|0.9|4|2|0.55|0.5| 0.5| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.55

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8999

FWER (P reject ≥ 1 | H0) : 0.0461

Max sample size (planned) : 3120

ESS under H1 : 2261.3

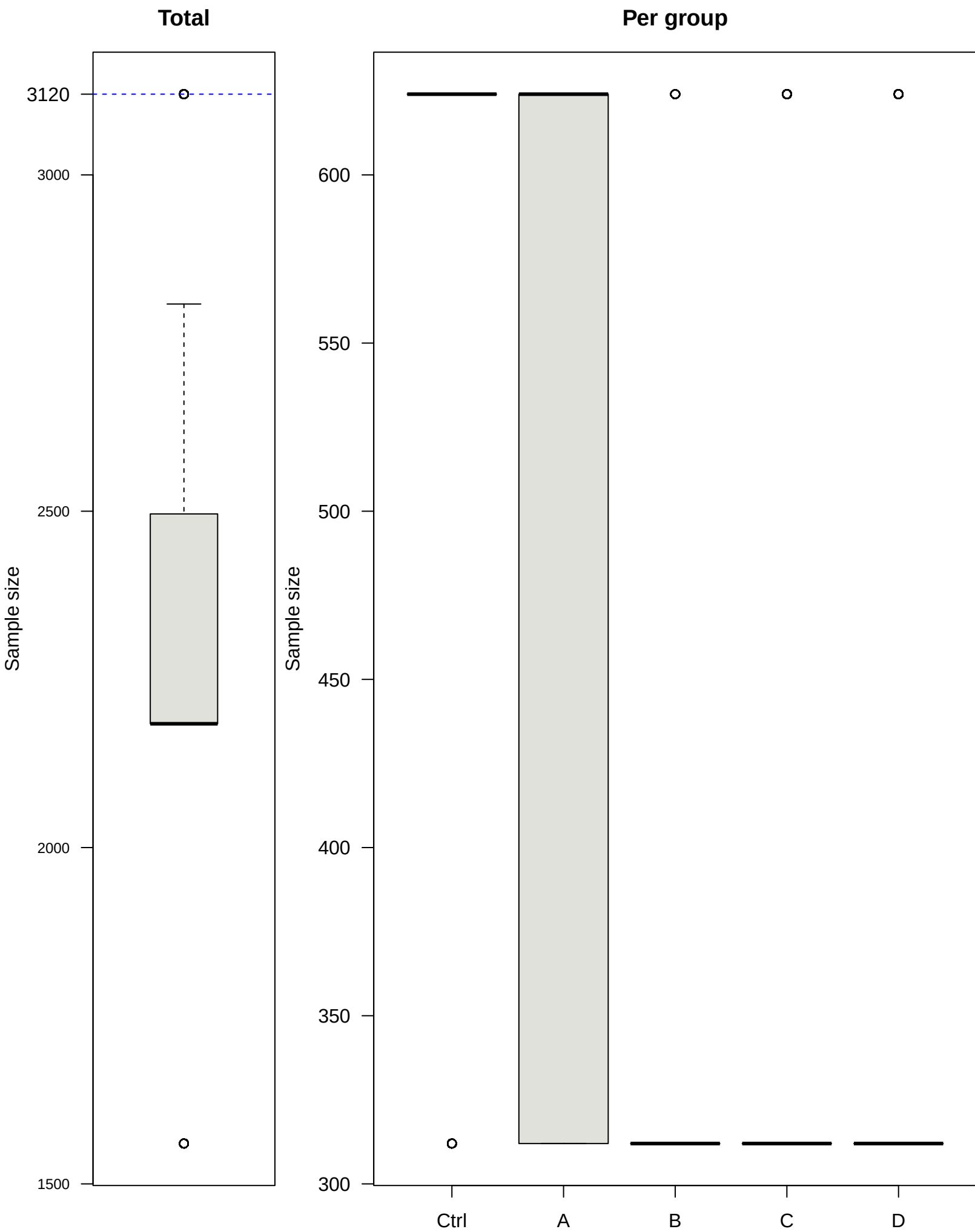
ESS under H0 : 1987.6

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 272 – sample size (H1) [binary]



Config 286 (0.05|0.9|4|2| 0.6|0.5| 0.7| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.6

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.700

effect pi_t = 0.900 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8814

FWER (P reject ≥ 1 | H0) : 0.0472

Max sample size (planned) : 500

ESS under H1 : 368.6

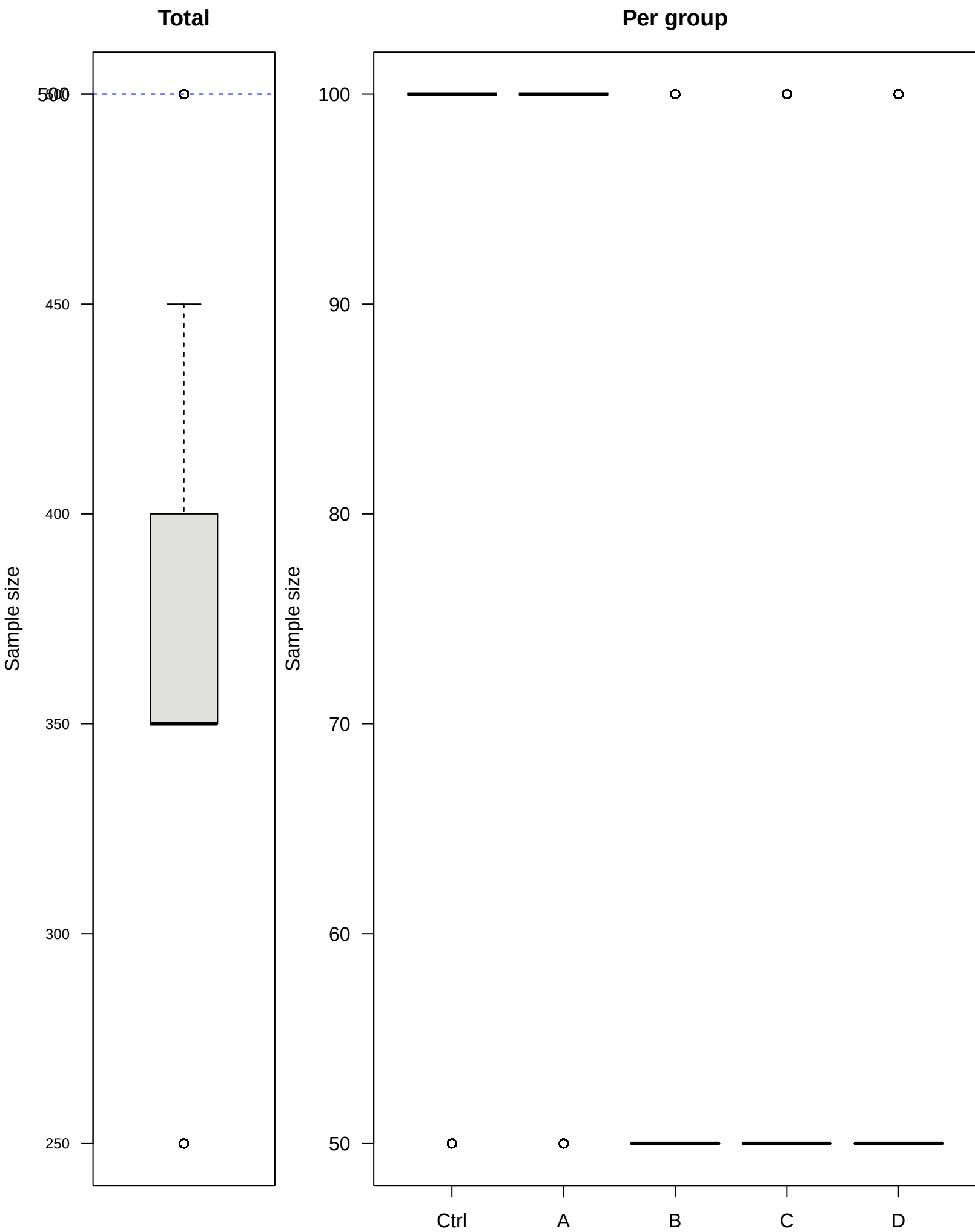
ESS under H0 : 318.7

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 286 – sample size (H1) [binary]



Config 290 (0.05|0.9|4|2|0.65|0.5|0.05| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.350$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9654

FWER (P reject ≥ 1 | H0) : 0.0521

Max sample size (planned) : 220

ESS under H1 : 142.1

ESS under H0 : 140.5

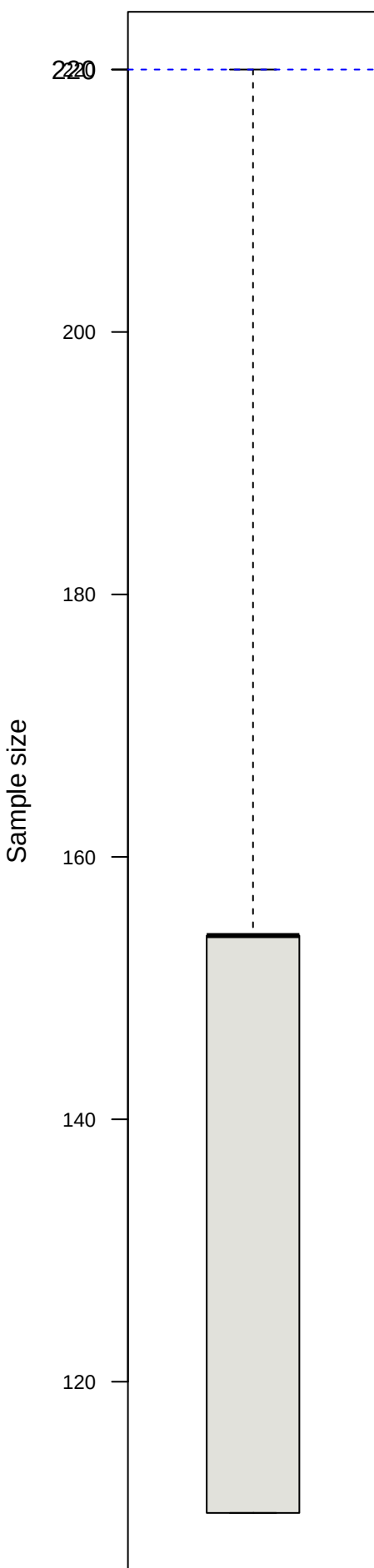
NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

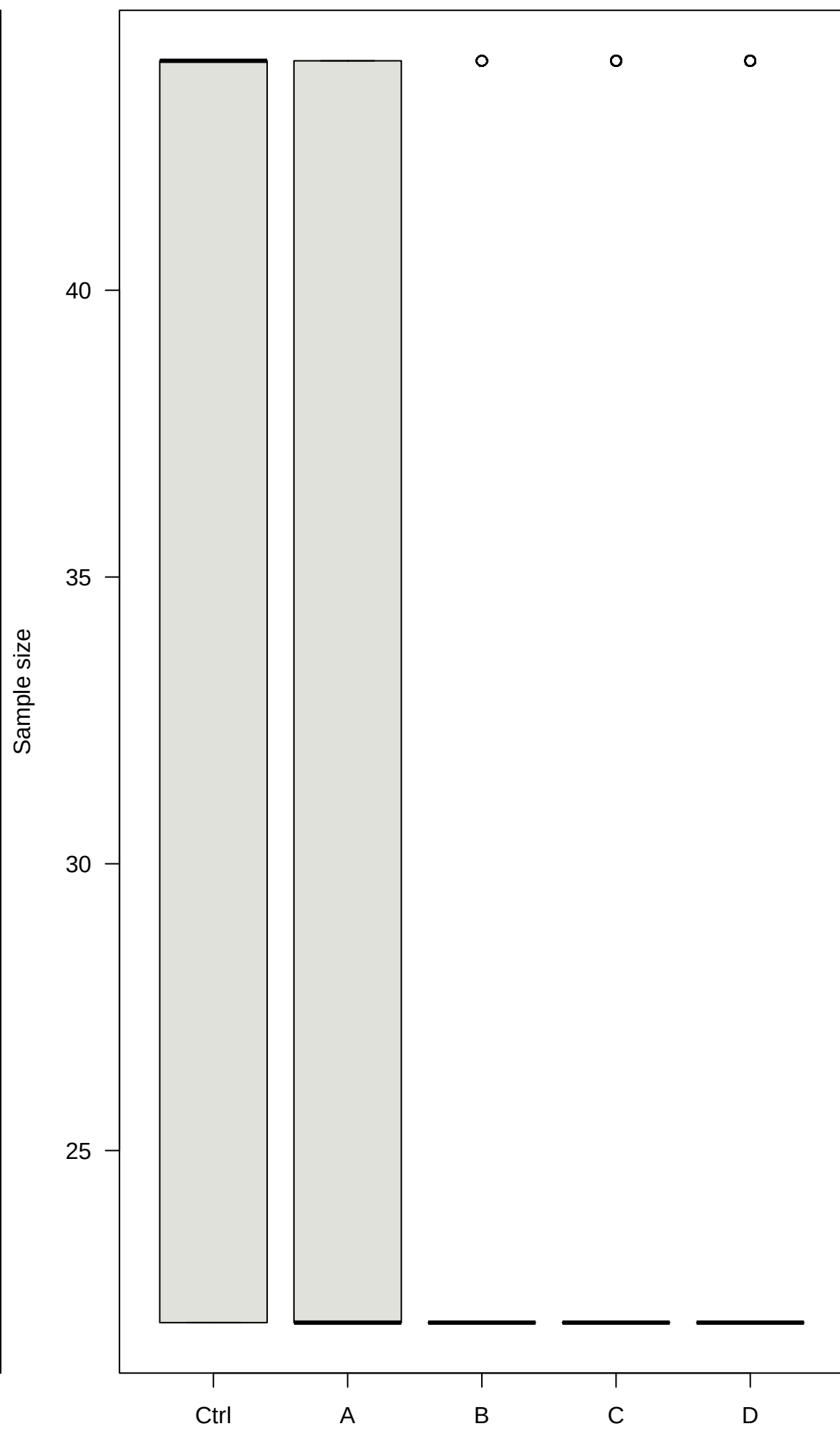
(delta/sd); residual gap from target is small-sample discreteness.

Config 290 – sample size (H1) [binary]

Total



Per group



Config 294 (0.05|0.9|4|2|0.65|0.5| 0.3| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.600 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9066

FWER (P reject ≥ 1 | H0) : 0.0496

Max sample size (planned) : 340

ESS under H1 : 241.5

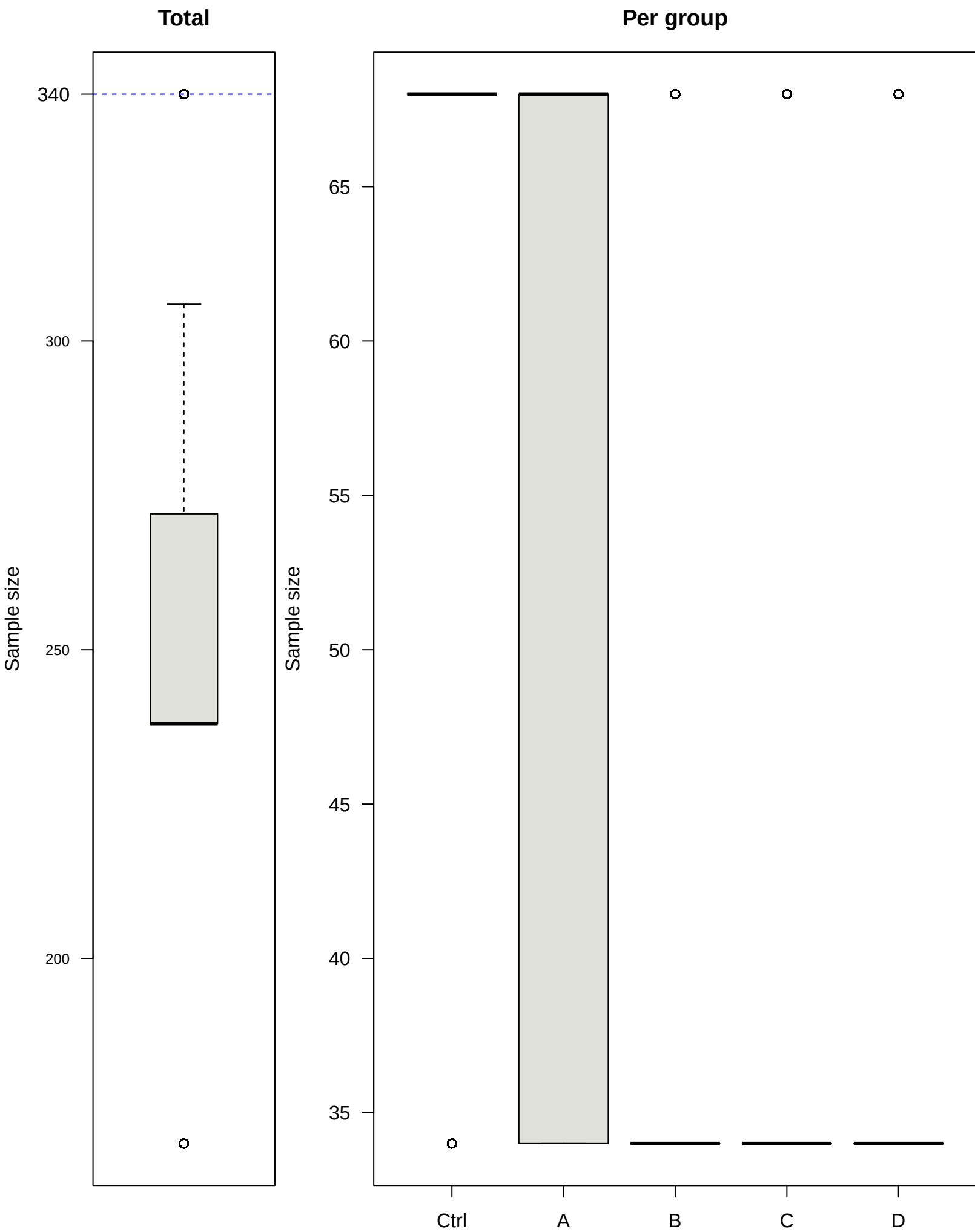
ESS under H0 : 220.5

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 294 – sample size (H1) [binary]



Config 296 (0.05|0.9|4|2|0.65|0.5| 0.5| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.65

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.500

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.8943

FWER (P reject ≥ 1 | H0) : 0.0389

Max sample size (planned) : 310

ESS under H1 : 227.1

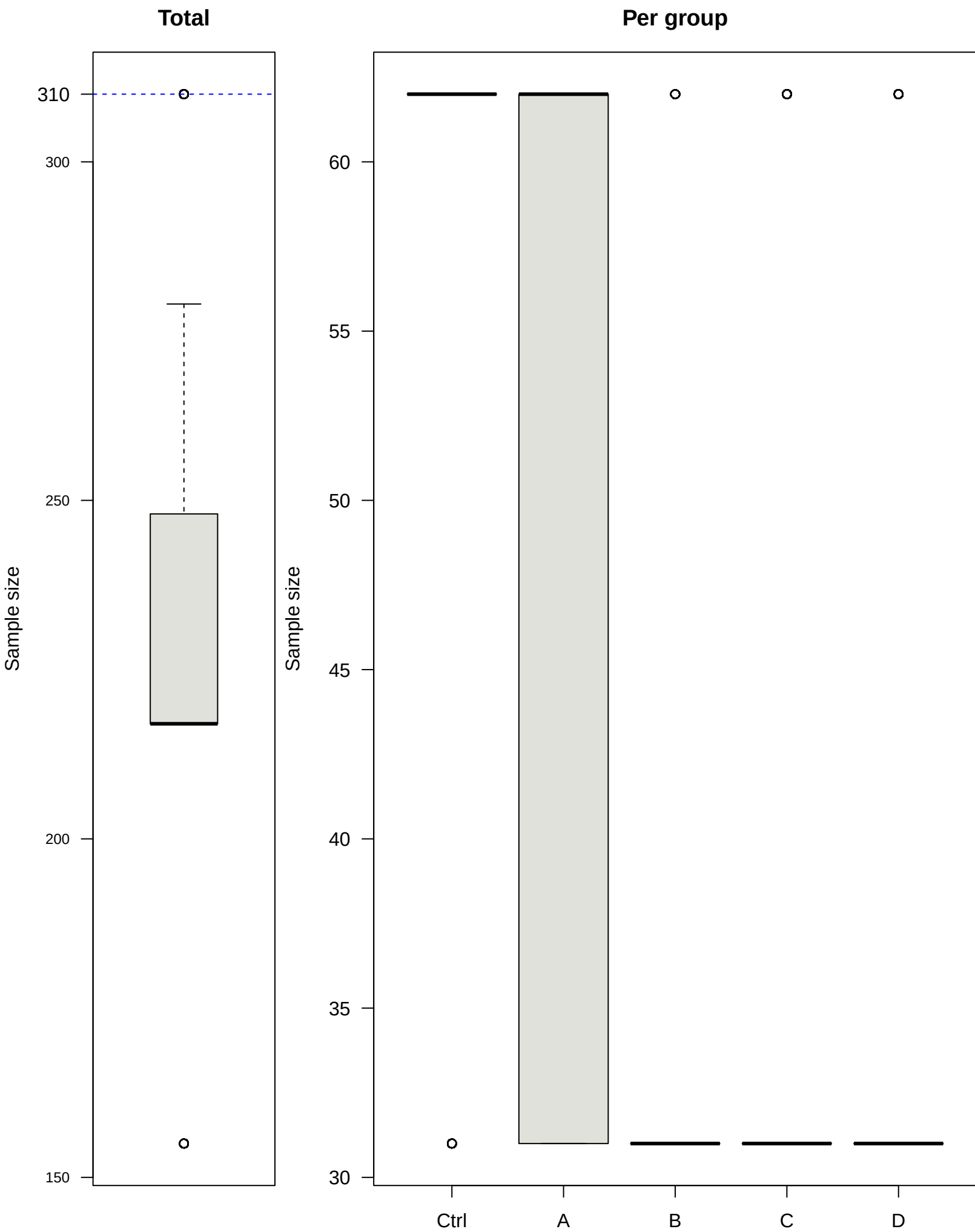
ESS under H0 : 192.9

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 296 – sample size (H1) [binary]



Config 306 (0.05|0.9|4|2|0.75|0.5| 0.3| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.75

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.300

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9239

FWER (P reject ≥ 1 | H0) : 0.0503

Max sample size (planned) : 120

ESS under H1 : 88.9

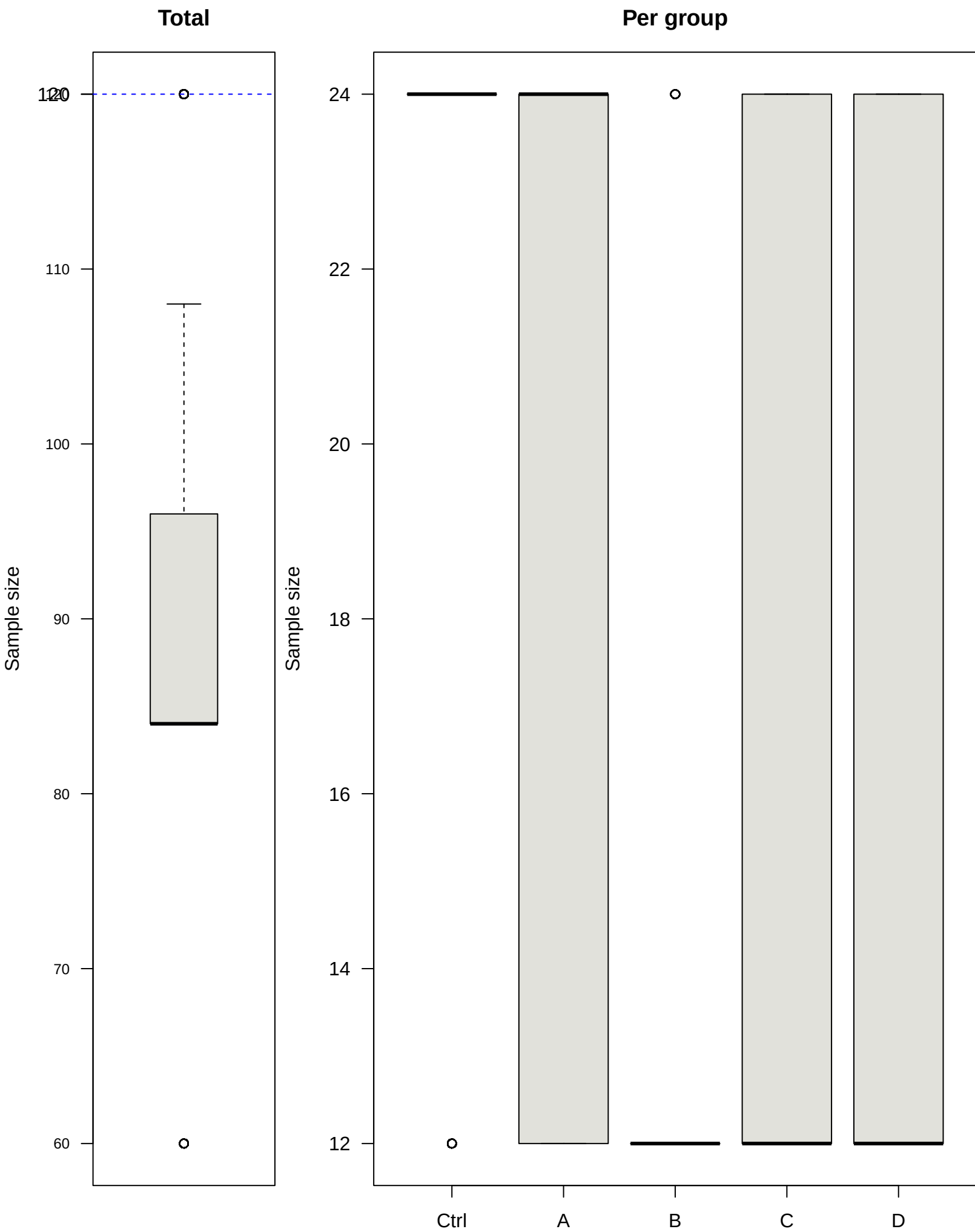
ESS under H0 : 78.3

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 306 – sample size (H1) [binary]



Config 316 (0.05|0.9|4|2|0.85|0.5| 0.1| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.85

p0 (null win-prob) : 0.5

method : sep

true response rates : control pi_c = 0.100

effect pi_t = 0.800 (= pc + 2p-1)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9331

FWER (P reject ≥ 1 | H0) : 0.0524

Max sample size (planned) : 50

ESS under H1 : 33.9

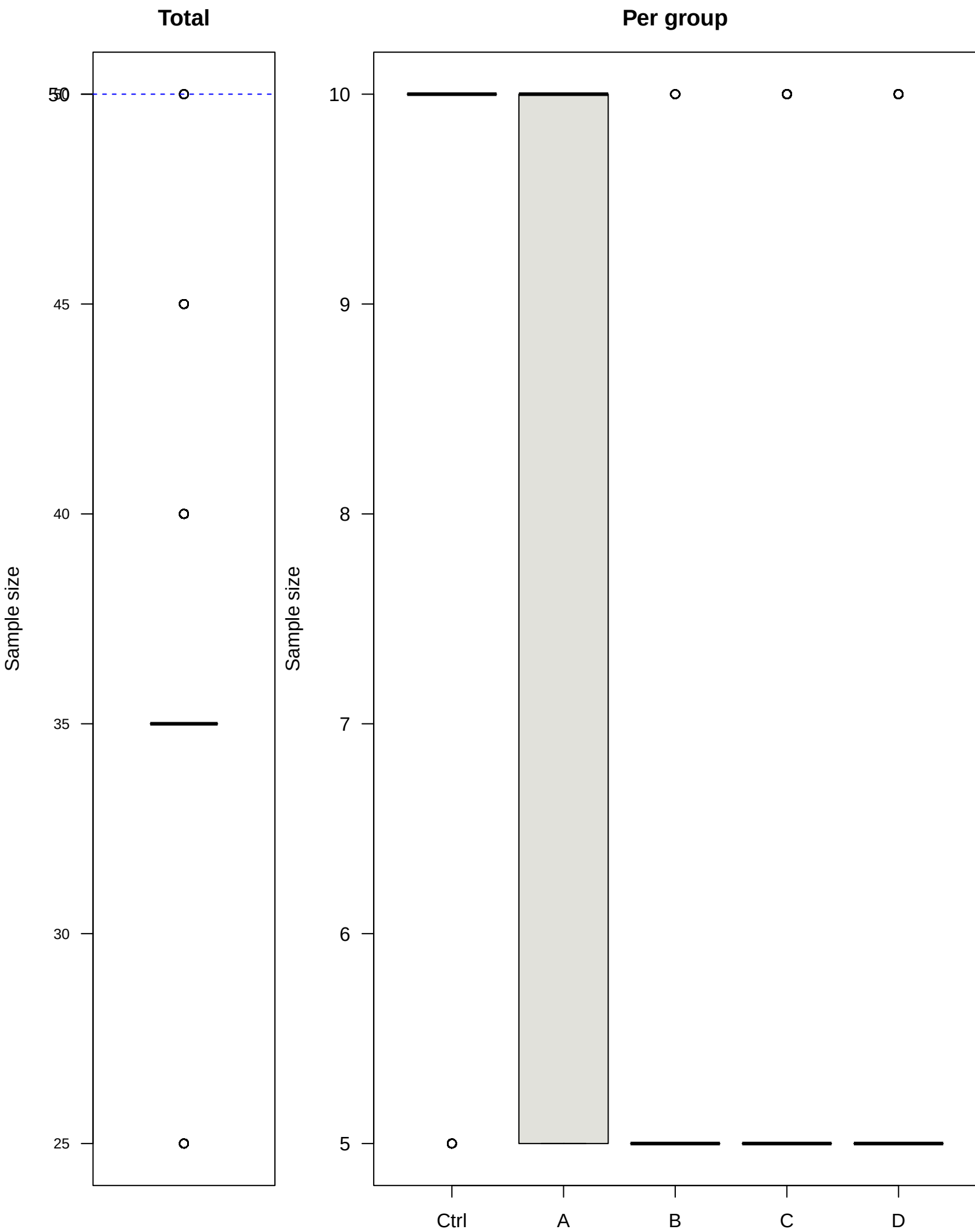
ESS under H0 : 35.0

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 316 – sample size (H1) [binary]



Config 326 (0.05|0.9|4|2|0.95|0.5|0.05| sep) [BINARY, NON-BINDING, FUT ENFORCED]

K (treatment arms) : 4

J (looks) : 2

p (target win-prob) : 0.95

p0 (null win-prob) : 0.5

method : sep

true response rates : control $\pi_c = 0.050$

effect $\pi_t = 0.950$ (= $\pi_c + 2p-1$)

target alpha = 0.05 target power = 0.9

SIMULATED OPERATING CHARACTERISTICS (Bernoulli, stats.ptest)

Power (P reject ≥ 1 | H1) : 0.9818

FWER (P reject ≥ 1 | H0) : 0.0140

Max sample size (planned) : 30

ESS under H1 : 22.1

ESS under H0 : 24.6

NB: non-binding (wider) boundary with futility ENFORCED, so power

and ESS are realistic. Design powered for the binary test

(delta/sd); residual gap from target is small-sample discreteness.

Config 326 – sample size (H1) [binary]

